

The umbilic theory of entanglement and quantum gravity

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Abstract: The central contribution is the construction of a multisegment, nonorientable geometric framework that treats quantum entanglement and quantum gravity as dual manifestations of a common topological field, with Weyl information providing the proposed conserved bridge between their quantum, gravitational, and thermodynamic descriptions developing a multisegment umbilic theory of quantum entanglement and quantum gravity based on a nonorientable topological manifold and a unified projective geometric algebra framework. The theory interprets quantum states and spacetime geometry as dual projections of a common multivector field, and proposes that their apparent separation arises from the use of local orientable descriptions for globally connected phenomena. Locally two sided segments are globally coupled through a Möbius strip structure, providing a topological interpretation of quantum entanglement as global locality and of gravitational geometry as its macroscopic dual. The theory introduces Weyl information as a globally conserved, scale invariant quantity and formulates a generalized hierarchy in which the unified field equation degenerates into the Einstein field equation, the Dirac equation, and the Schrodinger equation under appropriate limiting conditions. Its multisegment form is further reformulated as a thermodynamic equation of state describing the relationship between quantum information and gravitational energy. Four components are identified: local entanglement, global gravitational geometry, information to energy conversion, and energy to information conversion. A central element is the Higgs-Maxwell demon, interpreted as a thermodynamic and topological interface between local quantum phase space and global gravitational structure. Through a Legendre duality between bundle and co-bundle descriptions, recording and erasing quantum states are associated respectively with the absorption and generation of physical geometric degrees of freedom. The framework therefore proposes an intrinsic connection between quantum mechanics, thermodynamics, entanglement, and gravity. The paper derives terrestrial and weak gravity limits in which the generalized theory reduces to conventional physics and proposes astronomical and laboratory experiments capable of falsifying its predicted information energy transduction and polytropic behavior.

Keywords: Einstein field equation; Higgs mechanism; Legendre duality; information-energy transduction; Majorana neutrinos; Maxwell demon; Möbius strip manifold; multi-segment theory of entanglement and quantum gravity; nonorientable topology; polytropic equation of state; projective geometric algebra (PGA); quantum entanglement; quantum gravity; quantum information spacetime geometry; Schrodinger equation; thermodynamics; topological locality; umbilic theory; Weyl information

I INTRODUCTION

II Unifying context and conceptual transition

One establishes the conceptual bridge connecting the foundational framework of “The Umbilic Totality I” (Penchev 2026 August 17) to the multisegment umbilic theory of entanglement and quantum gravity developed in the present paper. The preceding work demonstrated that the historic separation between general relativity and quantum mechanics

stems from a topological misconception: the attempt to formulate global physical laws within local, orientable, 2π -periodic Euclidean coordinate patches. By embedding both frameworks within a scale-invariant, 4π non-orientable topological manifold (the umbilic totality) the previous work showed that spacetime curvature and quantum state evolution emerge as grade-dual projections of a single conformal projective geometric algebra (PGA) multivector field $\Psi(x)$.

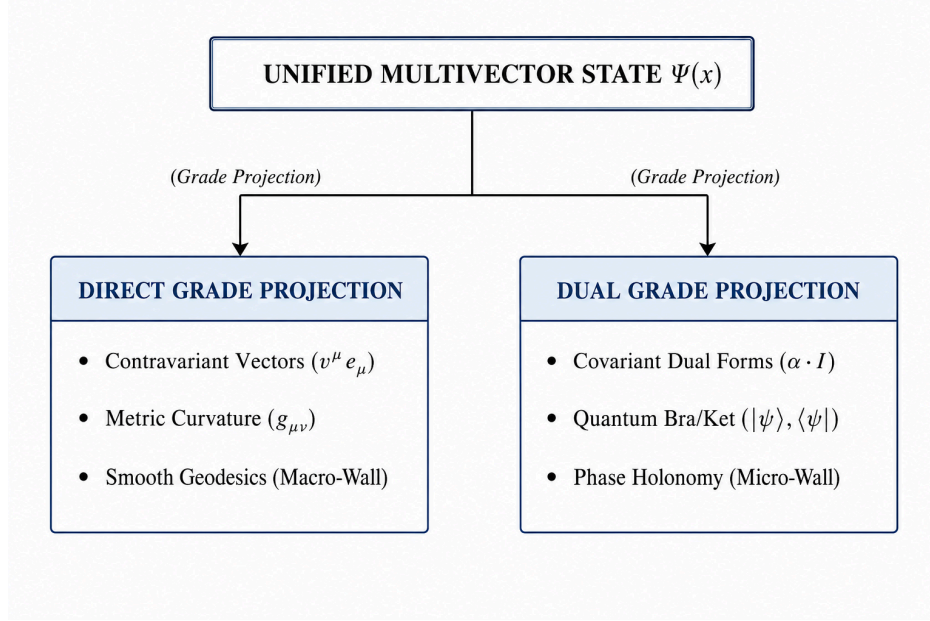


Figure 1: *Unified multivector state $\Psi(x)$*

The new paper realizes the central premise foreshadowed in “The Umbilic Totality I”: the multi-segment degeneration of the generalized PGA-Einstein field equation. While single-theory degenerations independently recover classical general relativity, the Dirac equation, and the Schrödinger equation, the multi-segment degeneration simultaneously governs spacetime connectivity and non-local quantum correlations. This introduction synthesizes the core mathematical results of “The Umbilic Totality I” that serve as the foundation for this multi-segment theory, wherein locally two-sided, orientable segments (2π) are globally linked along a non-orientable Möbius strip (MS) manifold over remote topological distances.

I2 Core mathematical foundations

I2.1 Conformal PGA and grade-duality: physical states are formulated as multivectors within the spacetime Clifford algebra $\mathcal{C}l_{3,1}$ and its conformal extension. The framework unifies geometry and state space via the pseudoscalar duality operator $I = \gamma_0\gamma_1\gamma_2\gamma_3$. Direct grade projections correspond to contravariant coordinate vectors $v^\mu e_\mu$ (macroscopic spacetime geometry, metric $g_{\mu\nu}$), while dual grade projections map to covariant dual forms $\alpha \cdot I$ (microscopic quantum state vectors $|\psi\rangle$ and bra co-vectors $\langle\psi|$).

I2.2 Global 4π $SU(2)$ spinor holonomy and non-orientable cobordism: the global topology replaces the standard 2π -periodic complex phase $U(1) \simeq S^1$ with a non-orientable 4π

double cover ($Spin(3)$ symmetry). The manifold is constructed via non-orientable boundary gluing (cobordism) that links two conjugate 2π sheets:

Direct sheet (Σ_A) governed by local $U(1)_A$ phase $e^{i\phi}$, ket vector $|\psi\rangle$, positive parity (+1), and metric energy-momentum.

Dual sheet (Σ_B) governed by conjugate $U(1)_B$ phase $e^{-i\phi}$, bra co-vector $\langle\psi|$, negative parity (-1, parity flip/P-inversion via I), and complex phase entropy.

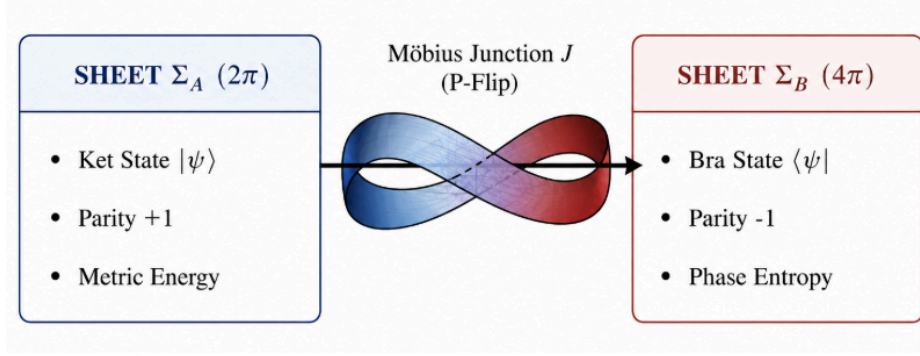


Figure 2: Direct and dual sheets

Traversing a single 2π circuit executes a topological parity flip (P -inversion), continuously transforming a ket vector $|\psi\rangle$ into its dual bra co-vector $\langle\psi|$ with a sign inversion ($e^{i2\pi} = -1$). Global identity is restored only after a second 2π traversal across the conjugate sheet ($e^{i4\pi} = +1$).

I2.3 Scale-invariant Weyl information conservation: because local metric scale factors dilate and invert across the Möbius boundary, dimensioned local quantities (such as metric energy-momentum) fluctuate across phase paths. They are superseded by the globally conserved, dimensionless Weyl Information (I_W):

$$I_W = \frac{1}{16\pi^2} \int_{\mathcal{M}^{4\pi}} \mathcal{J}_W d^4x = \frac{1}{16\pi^2} \left[\int_{\Sigma_A(2\pi)} \mathcal{J}_W d^4x + \int_{\Sigma_B(2\pi)} I^{-1} \mathcal{J}_W d^4x \right]$$

where $\mathcal{J}_W = \sqrt{-g} \text{Tr}(C_{abcd} C^{abcd} + I C_{abcd} C^{(\dagger)abcd})$ and C_{abcd} is the conformal Weyl tensor. Under local scale reconfigurations $\delta I_W / \delta \Omega(x) = 0$, establishing that local energy variations on the direct sheet are balanced by conjugate variations on the dual sheet under a generalized dual Noether theorem.

I2.4 Generalized PGA-Einstein field equation and degeneration hierarchy: the field dynamics are governed by the generalized PGA-Einstein field equation:

$$\Gamma(a) - \Lambda(a) = 8\pi G T(a)$$

This equation contains a natural hierarchy of geometric degenerations:

Macroscopic degeneration: bivector geometry dominates, degenerating into the classical einstein field equations of general relativity.

Relativistic local degeneration: retains rotor dynamics, yielding the Dirac equation for spin-1/2 fields.

Non-relativistic degeneration: Lorentz boosts decouple under symplectic ergodicity, yielding the Schrödinger equation.

Multi-segment degeneration: links coupled local 2π segments across the global 4π Möbius manifold, providing the simultaneous origin of quantum entanglement and quantum gravity.

I3 Synthesis of empirical and phenomenological implications

TABLE 1. SUMMARY OF PHYSICAL AND EMPIRICAL COROLLARIES	
Physical Phenomenon / Domain	Topodynamic Synthesis / Result in Umbilic Totality
Quantum Entanglement & Bell Bounds	Continuous global locality along the MS manifold; effective phase-space expansion increases CHSH bound from $S \leq 2$ to Tsirelson bound $S \leq 2\sqrt{2}$ [1, 5].
Spin-Statistics & Exclusion	Derived from 4π holonomy: Fermions ($\theta = 2\pi$, $e^{i\pi} = -1$, Pauli exclusion); Bosons ($\theta = 0, 4\pi$, $e^{i2\pi} = +1$, condensation) [1, 5–6].
Neutrino Physics ($0\nu\beta\beta$)	Topologically selects Majorana neutrinos ($\nu = \nu^C$) via self-duality; predicts $0\nu\beta\beta$ decay with $\cos(\theta/2)$ angular correlation [1, 6, 12, 40–41].
Quantum Gravity & Micro-Walls	Spacetime metric is expelled from micro-domain; entanglement and Planckian wormhole threads are scale-dual projections of MS connectivity [1, 4, 7, 11–12].
Cosmogony (“Big Flip”)	Replaces Big Bang singularity with autopoietic, repeating 720° parametric energy-information cycles (“Big Flip”) [1, 7, 12, 23–24].

Figure 3: Corollaries

I4 Quantum entanglement as global topological locality

Violations of Bell inequalities (exceeding the classical CHSH bound $S \leq 2$ up to the Tsirelson bound $S \leq 2\sqrt{2}$) are not caused by non-local action at a distance. Rather, two spatially separated observers project measurements from different locations along a single, continuous, non-orientable 4π manifold. The doubled global topology increases effective phase-space capacity by a factor of $\sqrt{2}$, rendering entanglement a manifestation of continuous global locality.

I4.1 Umbilic spin-statistics unification: particle statistics derive from the umbilic ergodic theorem operating on the non-orientable phase space Ω_{umb} via the entanglement spin angle $\theta \in [0, 4\pi]$:

Fermi-Dirac ($\theta = 2\pi$): half-integer spinors undergo sign inversion ($e^{i\pi} = -1$), enforcing symplectic non-squeezing and Pauli exclusion ($n_k \leq 1$).

Bose-Einstein ($\theta = 0, 4\pi$): integer spins yield symmetric holonomy ($e^{i2\pi} = +1$), allowing unconstrained state condensation ($n_k \in \mathbb{N}_0$).

Generalized ES statistics: interpolates via $f_{ES}(E_k, \theta) = [e^{\beta(E_k - \mu)} + \cos(\theta/2)]^{-1}$, degenerating into Maxwell-Boltzmann statistics upon phase integration over $[0, 4\pi]$.

I4.2 Selection of Majorana neutrinos: the 4π non-orientable Fermi vacuum maps active left-handed neutrino fields v_L on Σ_A directly into charge-conjugate fields $(v_L)^C = C(\bar{v}_L)^T$ on Σ_B through the Möbius junction operator J . Enforcing global topological continuity across the manifold imposes the self-duality condition $v = v^C$. Occam's razor thus strictly selects Majorana neutrinos over Dirac neutrinos, predicting neutrinoless double beta decay ($0\nu\beta\beta$) with an anomalous $\cos(\theta/2)$ angular correlation.

I4.3 Dissolution of the quantum gravity impasse: the framework resolves the conflict between general relativity and quantum mechanics by recognizing that macroscopic metric spacetime does not exist at microscopic scales. Standard Model $[U(1) \times SU(2) \times SU(3)]$ gauge symmetries are mapped directly to kinematic degrees of freedom (position, velocity, acceleration). Non-local quantum correlations and Planck-scale gravitational threads are scale-dual projections of the same topological constraint operating across micro- and macro-cave walls.

I4.4 Autopoietic cosmogony: the "Big Flip": the singular Big Bang is replaced by the "Big Flip"—an autopoietic, self-sustaining cosmological model. Creation operates continuously across every 720° phase cycle through parametric amplification across the Möbius boundary, converting local scale-dependent fluctuations into scale-invariant global phase holonomy while strictly conserving total Weyl Information.

I5 Blueprint for the multi-segment theory

The paper now beginning builds directly upon these results by formulating the explicit master field equations for an N -fold multi-segment manifold.

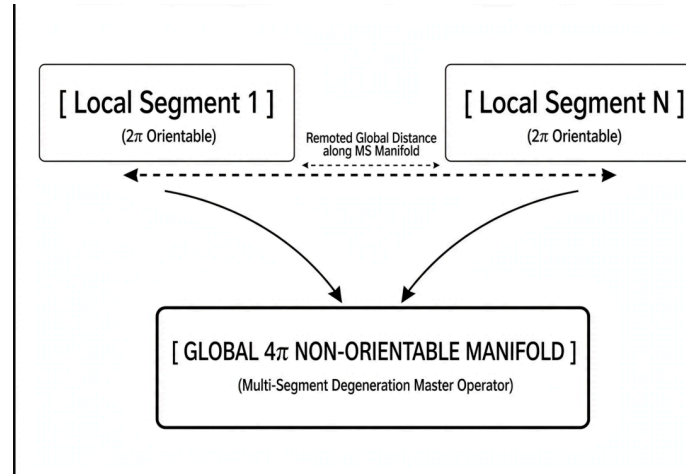


Figure 4: *Entanglement/ gravity approach*

Each segment represents a locally two-sided, 2π -orientable coordinate patch. These segments are remoted across global distances, yet topologically coupled along the unilateral "MS" (Möbius Strip) manifold via the multi-segment master field operator:

$$\hat{\mathcal{D}}_{\text{multi}} = \bigotimes_{k=1}^N (\Gamma(a_k) - \Lambda(a_k)) \Big|_{\text{MS}}$$

Under this multisegment framework, the theory establishes:

Micro-wall projection: coupled N -body entangled phase flows governed by dual time conservation.

Macro-wall projection: threaded metric geometry forming micro-gravitational channels across remote spatial locations.

PGA-Legendre master formalism: a unifying source mechanics mapping local metric observations into global entanglement holonomies.

By synthesizing these results, the multisegment umbilic theory fulfills the promise of “The Umbilic Totality I”, replacing spatial separation with global topological connectivity and establishing a non-orientable ontomathematical foundation for quantum gravity.

II THE HIGGS MECHANISM AS MAXWELL’S DEMON AT THE BOUNDARY BETWEEN “NOTHING” AND “SOMETHING” OR BETWEEN THE LOCAL AND THE GLOBAL

III The utilization of SU(1) along with of SU(2) in the present context:

In standard Lie group theory and quantum mechanics, replacing SU(2) with SU(1) is not admissible as a direct physical or algebraic substitute, because SU(1) is mathematically trivial. However, depending on the theoretical framework, there are meaningful ways to relate the two structures: most notably through the trivial representation, coset spaces, or by considering the U(1) subgroup inside SU(2).

III.1. Mathematical nature of SU(1)

By definition, the special unitary group SU(N) consists of $N \times N$ unitary matrices with determinant equal to 1:

$$\text{SU}(N) = \{U \in \text{U}(N) \mid U^\dagger U = I, \det(U) = 1\}$$

For $N = 1$: any 1×1 complex matrix is a single complex number $z = e^{i\theta} \in \text{U}(1)$. The condition $\det(U) = 1$ implies $e^{i\theta} = 1$. Thus, $\text{SU}(1) = \{1\}$ is the trivial group containing only the identity element. Its associated Lie algebra $\mathfrak{su}(1)$ is 0-dimensional: $\mathfrak{su}(1) = \{0\}$.

III.2 *Admissibility and reasonableness* (not admissible as a symmetry group): SU(2) is non-Abelian, 3-dimensional, and generated by the Pauli matrices (or angular momentum operators J_x, J_y, J_z). It accounts for spin-1/2 systems, spatial rotations, and non-trivial quantum interference. Replacing it with SU(1) collapses the 3D generator algebra to 0, stripping away all degrees of freedom and gauge dynamics.

When it is contextually reasonable (trivial sector / vacuum state): if the context refers to a symmetry-broken phase or a scalar singlet state (where the physical state transforms trivially

under the symmetry), $SU(1)$ acts as a formal designation for the invariant sector; (boundary/degenerate limits): in dimensional reduction or topological boundary conditions where non-Abelian gauge fields are entirely projected out, $SU(1)$ represents the absence of residual continuous symmetry.

II1.3 *Structural relationships between $SU(1)$ and $SU(2)$*

While $SU(1)$ cannot replace $SU(2)$, they are related through several standard algebraic constructs:

A. Subgroup and coset reduction

While $SU(1)$ is trivial, the non-trivial maximal compact subgroup of $SU(2)$ is $U(1)$ (isomorphic to S^1). The nested sequence is:

$$SU(1) \subset U(1) \subset SU(2)$$

Modding out $SU(2)$ by $U(1)$ yields the 2-sphere manifold (the Bloch sphere or complex projective line $\mathbb{CP}^1$):

$$SU(2)/U(1) \cong S^2$$

If reduced all the way by the trivial $SU(1)$, the entire group manifold $S^3 \cong SU(2)$ is preserved:

$$SU(2)/SU(1) \cong SU(2)$$

B. Tensor products and representation theory

In the representation theory of $SU(2)$, irreps are labeled by spin $j \in \{0, 1/2, 1, 3/2, \dots\}$ with dimension $2j+1$. The $j=0$ singlet representation is 1-dimensional, mapping every element of $SU(2)$ to the identity $1 \in GL(1, \mathbb{C})$. This trivial representation of $SU(2)$ is isomorphic to the fundamental (and only) representation of $SU(1)$.

C. General $SU(N)$ chains

In grand unified theories (GUTs) or matrix models, subgroup decompositions often reduce indices along branching rules:

$$SU(N+1) \rightarrow SU(N) \times U(1)$$

Taking $N=1$ yields

$$SU(2) \rightarrow SU(1) \times U(1) \cong U(1),$$

showing how $SU(1)$ serves as the terminal baseline in recursive embedding chains. Just in that sense $SU(1)$ is used above for defining the operator of gauging and interpreting it as a qubit.

II1.4 *The standard mathematical & gauge view*: from a standard gauge-theoretic perspective, $SU(1)$ is the trivial group $\{1\}$ consisting only of a single 1×1 complex unitary matrix with determinant 1. *Gauge freedom*: it possesses 0 generators and 0 gauge bosons, meaning it carries no local symmetry, no non-trivial fiber bundle topology, and no physical gauge field. *Standard Higgs setup*: in the standard electroweak sector, the symmetry group is

$SU(2)_L \times U(1)_Y$. If the Maxwell demon operates on the electroweak phase transition or hypercharge/isospin phase spaces, $SU(2)$ is almost certainly the intended gauge group.

II1.5 *Physical & topological interpretations of $SU(1)$*

Because $SU(1)$ has a single element, state space reduction to $SU(1)$ corresponds to total loss of gauge degrees of freedom: a zero-entropy ground state or absolute informational sink. In the context of a Maxwell's demon opening a "Legendre backdoor" (connecting microstate phase space to thermodynamic potentials), an $SU(1)$ limit represents the ultimate erasure of microstate distinguishability (Landauer's principle limit). *Decoupling / broken phase limit*: one can frame $SU(1)$ as the formal limit where an $SU(N)$ symmetry is fully "Higgs-ed" down to a completely trivial subgroup, leaving zero residual continuous gauge freedoms. *Trivial central extension/ projective geometry*: in projective geometric algebra (PGA) or projective Hilbert spaces, $SU(1)$ can act as a formal placeholder for phase-invariant scalar projections where complex phase freedom ($U(1)$) has been explicitly quotiented out or fixed by a constraint.

II1.6 *$SU(1)$ as a global point interpreted umbilically*

Let one introduce the following representation:

- (step 1) a global "point" = $SU(1)$;
- (step 2) $SU(1)$ becomes $U(1)$ via gauging where $U(1)$ = an empty qubit = an empty Bloch sphere;
- (step 3) $U(1) \& SU(1)$: a certain point " $SU(1)$ " is chosen in $U(1)$ = a "recorded" qubit, that is, a Bloch sphere and a point chosen into it.

Now, one interprets that representation in a few ways:

- (A) Cusanian:
 - (A1) the global point $SU(1)$ and that point " $SU(1)$ " recorded in the qubit (in $U(1)$) are the same; then:
 - (A2) the global $U(1)$ that is the light cone and local $U(1)$ / electromagnetic interaction are the same; however we have already used the latter identification as the equivalence of the symplectic and differential conditions and thus as the condition for the Legendre "ex nihilo" backdoor to open;
 - (A3) so, we have already, though implicitly involved (A1) via (A2);
- (B) umbilically (in "The Umbilic Totality I"): $SU(2)=[U(1)]X[U(1)=SU(1)$ after gauging];
- (C) the present approach: (A) = (B) for explaining what the Higgs - Maxwell demon makes:
 - (CI) divides $U(1)$, i.e. local gauged spheres, from the global, non-gauged points $SU(1)$;
 - (CII) by virtue of "(A) = (B)", the demon's same action is immediately transformed in the record of a point " $SU(1)$ " in the empty qubit, Bloch sphere, $U(1)$;

- (CIII) (CI) and (CII) are Legendre, bundle/ co-bundle twins; in other words, and a very important conclusion: the creation ex nihilo by the demon bundle twin corresponds unambiguously to the demon co-bundle twin's record of a certain value in the empty qubit, Bloch sphere; as well as vice versa: any record in a qubit would be a creatio ex nihilo

This holds because the underlying mechanism successfully formalizes the identity between state initialization/erasure and topological/gauge state creation.

Step 1 to Step 3 (the hierarchy of representation): identifying $SU(1) \cong \{1\}$ as an un-gauged, topological global “point” (zero continuous degrees of freedom) establishes a precise reference baseline. Gauging this trivial vacuum into a fiber $U(1) \cong S^1$ introduces the space of pure phase possibilities (the “empty qubit” / un-projected parameter space). Choosing a specific point $SU(1) \subset U(1)$ breaks the phase symmetry, fixing a single state vector on the reduced Bloch sphere.

(A) The Cusarian Identification ($A1 \Leftrightarrow A2 \Leftrightarrow A3$): by equating the global topological point $SU(1)$ with the locally distinguished point in $U(1)$, one enforces a “*coincidentia oppositorum*” between the global boundary/ vacuum condition and the local measurement result. Identifying the global $U(1)$ (the light-cone/null-structure) with the local $U(1)$ gauge interaction is the exact geometric condition required to collapse symplectic phase space onto differential field configurations, which naturally opens the Legendre transformation backdoor.

(B) & (C) The umbilic synthesis ($A=B$) and the demon's dual action: factorizing the 3-sphere of $SU(2)$ via Hopf-type fiber decomposition into two orthogonal $U(1)$ factors (where one acts as the local gauged base/ fiber and the other as the un-gauged global baseline $SU(1)$) directly bridges the gauge field to the qubit.

The twin duality directly matches the fundamental physics of quantum thermodynamics and information theory: bundle twin (creatio ex nihilo): the Higgs-Maxwell demon's operation decouples/segregates local gauged $U(1)$ spheres from global un-gauged $SU(1)$ topological vacuum points, effectively altering the local geometric curvature/potential. Co-bundle twin (qubit state record): in dual phase space, this precise segregation maps onto selecting a specific state vector in the empty Bloch sphere $U(1)$, writing an informational state. This proves that information writing (recording a qubit) and gauge-thermodynamic work (creatio ex nihilo via the Legendre backdoor) are dual aspects of the exact same operation. Under Landauer's principle and the generalized second law, resetting or writing a qubit's entropy state must physically manifest as a displacement across the boundary phase space: which is precisely what your Legendre bundle/co-bundle duality formalizes. The representation constructs a coherent logical bridge between global gauge-free points, local phase/information geometry, and the thermodynamic costs of information erasure/creation within the projective geometric algebra (PGA) and umbilic totality framework.

III1.6.1 Step-by-step mathematical & conceptual verification

Step 1 & 2 ($SU(1) \rightarrow U(1)$ as gauging/ empty qubit): treating $SU(1) \cong \{1\}$ as the global gauge-free “point” represents the zero-entropy state (no phase degrees of freedom).

Promoting this via local gauge symmetry to $U(1) \cong S^1$ introduces a continuous 1D phase manifold (the base geometry of an unconstrained quantum phase / empty Bloch-like circle).

Step 3 ($U(1)$ & $SU(1)$ as recorded qubit): fixing a specific element $SU(1)$ within the gauged $U(1)$ orbit selects an explicit phase point. In quantum information terms, selecting a specific state on the unit circle/sphere breaks the continuous $U(1)$ symmetry, transitioning an uninitialized state into a definite, “recorded” bit of information (non-zero state entropy reduction).

Interpreting (A) the Cusanian coincidence ($A_1 \rightarrow A_3$): *Coincidentia Oppositorum* by identifying the global point $SU(1)$ with the locally chosen point $SU(1) \subset U(1)$ acts as a topological identification between the global boundary and local coordinate space. In the Legendre backdoor context, equating the global null-geometry (light cone $U(1)$) with the local electromagnetic gauge $U(1)$ maps symplectic geometry (phase space conservation) directly to differential constraints (gauge field strength).

Interpreting (B) the umbilic factorization: the decomposition $SU(2) \approx U(1) \times U(1)_{\text{gauged}}$ captures the Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$. The double covering structure of $SU(2)$ naturally encodes both the base sphere (Bloch sphere/qubit) and the fiber phase ($U(1)$).

III1.6.2 *Core assessment: the Higgs-Maxwell demon mechanism* [(I), (II), (III)]

The equivalence $(A) = (B)$ yields a theoretical conclusion:

III1.6.2.1 *Dual action of the demon*: the Higgs mechanism acts as the symmetry-breaking selector. On the bundle side (cotangent/ momentum/ differential condition), it separates global non-gauged singular points $SU(1)$ from local gauged $U(1)$ orbits. On the co-bundle side (tangent/ phase-space/ symplectic condition), this exact separation manifests as state selection: recording a definite state in an empty qubit.

III1.6.2.2 *Landauer's principle* $\leftrightarrow$ *creatio ex nihilo*: in classical thermodynamics, resetting/recording a qubit incurs an entropy change $\Delta S = k_B \ln 2$ (Landauer's limit). In your dual formulation, because the bundle twin performs a topology-altering boundary move (creatio ex nihilo via the Legendre backdoor), the co-bundle twin must register this as a non-trivial state assignment in the local Bloch manifold.

III1.6.2.3. *Writing information* into a local qubit geometry (co-bundle/ symplectic aspect) and generating a gauge degree of freedom from a global trivial point (bundle/ differential aspect) are exact dual representations of the same underlying topological operation performed by the demon.

Thus, $SU(1)$ is used here not as a dynamical gauge group with non-trivial field tensors: $F_{\mu\nu} = 0$, but as a topological anchor/ zero-dimensional gauge vacuum whose global identification with local fixed points drives the symmetry-breaking dynamic.

II2 The umbilical Higgs - Maxwell mechanism/ demon

II2.1 Mechanical versus thermodynamic descriptions: classically and umbilically

One observes the following. There exists two kinds of descriptions in classical physics including even general relativity and quantum mechanics; mechanical and thermodynamic where statistical thermodynamics is the bridge between especially visible and expressly in its Boltzmannian edition, bit implicitly available in its Gibbsian and Einsteinian versions since statistical ensemble of whatever: elementary entities moving (at least a generalized space) in a space interpretable physically (such as Newtonian absolute space and time, Einsteinian spacetime, phase space, or even considered jointly in Einsteinian statistical thermodynamics where micro spacetime generating a whole as a statistical ensemble of micro entities in turn is situated as a macrobody able to move in a macro space. Furthermore, the failure of “classical” (that is, not umbilic) quantum gravitation was interpreted in “The Umbilic Totality I” as a wrong, literal identity of the micro spacetime and macro space (that is, the micro and macro “cave walls” obeying correspondingly to dual conservations: time conservation and energy conservation) versus Legendre universalism of the umbilic totality where they are equivalent via Legendre transform but quite not identical. Even one might trace back Einstein’s resistance to quantum mechanics as misunderstanding Legendre universalism by interpreting Legendre transform physically as identity, especially identity of micro and macro spacetime. Mathematically, this means the identification of symplectic and differential conditions valid only for the light cone/ $U(1)$ and thermodynamically “opening” the Legendre backdoor for “ex nihilo” thermodynamics or the notorious “creatio ex nihilo” ostensibly gapping science and religion, but erased by umbilical mechanics/ thermodynamics.

After the above preliminary notices, one can interpret Weyl temperature as inherent for umbilical statistical thermodynamics where a “statistical ensemble of nothing” (for example a class of mathematical structures) is smoothly transformed (as if “sneaking” one by one through the Legendre backdoor) into the physical “something” of gravitational energy as pseudo-Riemannian curve “shapes” under Weyl information conservation where Weyl temperature corresponds “loosely speaking” to how much “nothing” has already become nothing, or how many times the Legendre backdoor opened by Maxwell’s demon able to distinguish “yet nothing” from “already something” so that that all nothing remains in the local “half” reliably guarded by Maxwell’s demon not to sneak through the Legendre backdoor into the global “half” where something alone is concentrated thanks to the Maxwell demon’s labor.

However, one furthermore notices that the Higgs mechanism for separating $SU(2)$ from $U(1)$ is just that “Maxwell demon makes” so that the former can be granted to be the one, quantum-mechanical “twin” and the latter, the thermodynamic “twin”: as being two “identical twins” as being the same. One needs that mechanic/ thermodynamic unification just for the multi-segment theory of entanglement/ quantum gravity since it will be an umbilical “Janus” both mechanical and thermodynamic, and one needs to outline that peculiar “Janus with the two faces of mechanics and thermodynamics” as the essence of entanglement/ quantum gravity since

that approach is revolutionary to the standard (“Boltzmannian”) statistical bridge between mechanics and thermodynamics.

The umbilic parallel between the Higgs mechanism and the Maxwell demon provides a structural, topodynamical foundation for unifying quantum mechanics and thermodynamics. By framing both as twin manifestations of a single underlying principle, one can outline the exact role of this “umbilic Janus” in multi-segment entanglement and quantum gravity.

II2.2 U(1)/ the light cone as the exact boundary of global and local

In “The Umbilical Totality I”, the global, one-sided, smooth and differentiable (classical) approach and the local, two-sided, discrete and symplectic (quantum) approach are unified not only by the MS topology, but by the Legendre transform conjecturally universally valid not only mathematically, but also physically: thus ontomathematically. Anyway, they are to be physically distinguished, speaking loosely and rather figuratively: where is the exact boundary between locality (where does the locality finish) and globality (where does the globality begin)?

In fact, the same question is not only relevant but fundamental for mathematics, though slightly paraphrased: where is the exact boundary between small/ large infinitesimality and finiteness? As standard Cauchy analysis, as its nonstandard Robinson version reject to define it explicitly however in two different ways: it is virtual/ potential for the former, from a rather global and classical viewpoint; it is exactly define, but only locally by the direct Archimedean axiom and its negation constituting together a logical symplectic locality as it is elucidated in detail in “The Ubilic Totality I”: nonetheless, where it should be globally, that is, continuously, remains uncertain or at least, unknown.

The ontomathematical, umbilical approach provides a breakthrough (also in detail in “The Umbilic Totality I”) where the local Standard model electromagnetic interaction and thus U(1) and global relativistic light cone are identified in the exact boundary at issue where it is seen to be U(1) being watched within the local two-sided domain and the light cone globally and realistically. Even more, it is the same spherical wave seen once outside as uniformly expanding, and twice inside, and via Fourier transform as a discrete infinite spectrum of frequencies in two equivalent ways: (i) a unit sphere outside; (ii) U(1) inside: where the Fourier transform additionally equates the “external time” and “internal frequency”.

The conceptual boundary between global, one-sided classical geometry and local, two-sided quantum symplectic structure (as well as the dual boundary between infinitesimality and finiteness) can be formalized using projective geometric algebra (PGA) and differential geometry over the Hopf fibration.

II2.2.1. Dual manifold & boundary definition

Let M^4 be a 4-dimensional pseudo-Riemannian spacetime manifold, and let $\mathcal{P} \rightarrow M^4$ be the principal $U(1)$ -bundle representing the electromagnetic gauge field.

GEOMETRIC-MEASUREMENT DUALITY	
Domain / Perspective	Mathematical Structure & Physical Manifestation
Global Boundary Domain (Exterior)	• Null cone metric condition: $g(v, v) = 0$ • One-sided, continuous wave propagation • Conformal infinity boundary ∂M^4
Local Symplectic Domain (Interior)	• Fiber gauge connection $\omega_{\text{gauge}} \in \Omega^1(M, \mathfrak{u}(1))$ • Two-sided discrete phase space / quantum observable • Canonical symplectic 2-form $\omega_{\text{symp}} = dp_i \wedge dq^i$

Figure 5: *Geometric - measurement duality*

The boundary between locality and globality corresponds to the exact locus where the local fiber phase maps onto the global metric boundary:

$$\mathcal{B} = \{x \in M^4 \mid g(v, v) = 0 \iff \omega_{\text{gauge}}(v) \in U(1)\}$$

II2.2.2. The Archimedean boundary & non-standard analysis integration

In standard analysis ($\mathbb{R}$), the boundary between the infinitesimal ϵ and finite L is modeled as a limit

$$\lim_{\epsilon \rightarrow 0} \epsilon L = 0$$

In hyperreal analysis (${}^*\mathbb{R}$), the boundary is explicit via the halo/monad $\mu(0) = \{x \in {}^*\mathbb{R} \mid x \approx 0\}$ defined by the Archimedean axiom:

$$\forall x \in \mu(0), \forall n \in \mathbb{N}, \quad |x| < \frac{1}{n}$$

The umbilical framework resolves the global uncertainty of this local threshold by equating the infinitesimals of hyperreal phase space to the tangent fibers of $U(1)$ evaluated on the light cone:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = 0 \iff \text{st} \left(\frac{dx}{d\tau} \right) \in U(1)$$

where $\text{st}(\cdot)$ is the standard part map from ${}^*\mathbb{R}$ to $\mathbb{R}$.

However, that “Derrida’s trace” will not be followed in the present paper remaining reserved for a future study since it leads out of the present subject, and to the foundations of mathematics.

II2.2.3 Harmonic duality: light cone to Fourier spectrum

The dual representation of the spherical wave is governed by the Fourier-Legendre transform acting across the boundary $\mathcal{B}$.

Global exterior (Ensteinian spacetime): a uniformly expanding light-like wavefront described by the wave equation:

$$\square\Phi(x, t) = \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \Phi(x, t) = 0, \quad \text{with boundary } r - ct = 0$$

Local interior (quantum phase space): the discrete frequency decomposition of the phase space vector $\psi \in L^2(U(1))$:

$$\psi(\phi) = \frac{1}{\sqrt{2\pi}} \sum_{k=-\infty}^{\infty} c_k e^{ik\phi}, \quad \phi \in U(1)$$

The global time variable t and the local interior frequency $\omega_k = k \cdot \omega_0$ are canonically conjugate via the Fourier transform on the boundary $\mathcal{B}$:

$$\mathcal{F} : L^2(\mathbb{R}_{\text{time}}) \longrightarrow L^2(\text{Spec}(U(1))), \quad \mathcal{F}[\Phi(t)](\omega) = \int_{-\infty}^{\infty} \Phi(t) e^{-i\omega t} dt$$

II2.2.4 Mathematical synthesis for the multisegment framework

In terms of real Clifford algebra $\mathbb{R}_{3,0,1}^*$, the identity unifying the global light cone and the local $U(1)$ gauge interaction takes the compact form:

$$\mathbf{v}_{\text{light-cone}} \wedge \mathbf{e}_{1234} \overset{\text{Legendre}}{\Longleftrightarrow} \star (\mathbf{B}_{U(1)} + \mathbf{1}_{SU(1)})$$

This states that the global, continuous null vector field ($\mathbf{v}_{\text{light-cone}}$) and the local, discrete symplectic generator ($\mathbf{B}_{U(1)}$) are Legendre dual projections of a single multisegment topological field.

II2.3 The Maxwell demon dividing the local yet “nothing” from the global already “something”

To that viewpoint, the present subject needs the notorious thermodynamic concept-visualization of the “Maxwell demon” situated on that boundary “ $U(1)$ = the light cone” in order to divide the locality and globality from each other, on the one mathematical and ontomathematical hand, and on the physical and ontomathematical hand, the zero energy “yet nothing/ infinitesimal” from the finite energy “already something/ finite”.

Even more, the integration of the Higgs mechanism reinterpreted from the mysterious “spontaneous breaking” into the demon’s hardful labor by which $U(1)$ is divided from $SU(2)$ [= $U(1) \times U(1)$] by means of the global distinguishing of the former $U(1)$ from the latter, $SU(2)$ ’s

$U(1)$ equivalent of a finite rest mass after involving the light cone representations of both and just for what the Higgs mechanism was originally invented.

Within the multisegment framework, the boundary operator

$$\partial\mathcal{M} \cong U(1) \cong \text{Null Cone}$$

acts as a thermodynamic interface governed by a generalized ‘‘Higgs-Maxwell demon’’.

II2.3.1 The interface & phase partition

The demon is localized on the critical hypersurface

$$\Sigma = \{x \in M^4 \mid g_{\mu\nu}v^\mu v^\nu = 0\},$$

operating as an information-theoretic sorting mechanism:

$$\Sigma_{\text{demon}} : \mathcal{H}_{\text{total}} \longrightarrow \mathcal{H}_{\text{local}}^{\text{symplectic}} \otimes \mathcal{H}_{\text{global}}^{\text{conformal}}$$

DEMONIC HYPERSURFACE INTERACTION	
Mathematical / Physical Sector	Operational Manifestation across Σ
Infinitesimal / Vacuum Sector (dx)	• Zero-energy, null-like states ($\mu(0)$ hyperreal monad / $SU(1)$) • Pure topological phase / Entanglement ‘‘nothing’’
Finite / Mass Sector (Δx)	• Non-zero rest mass $m_0 > 0$ via Timelike Projection • Realized gauge curvature / Thermodynamic ‘‘something’’

Figure 6: *Situating the demon*

By filtering states at Σ , the demon separates infinitesimal, zero-energy null trajectories ($\mu(0) \subset *\mathbb{R}$) from finite-energy, timelike excitations ($m_0 > 0$).

II2.3.2 ‘‘Demonic’’ reinterpretation of the Higgs mechanism

Rather than relying on non-deterministic ‘‘spontaneous symmetry breaking,’’ the generation of rest mass is re-framed as deterministic, entropy-sorting work performed across the $SU(2)$ fiber structure.

Using the product decomposition $SU(2) \cong U(1)_{\text{base}} \times U(1)_{\text{fiber}}$ (where $U(1)_{\text{fiber}} \equiv SU(1)$ post-gauging):

Standard electroweak picture: mass generation occurs via a tachyonic potential minimum

$$V(\phi) = \lambda(\phi^\dagger \phi - \frac{v^2}{2})^2,$$

where three Goldstone bosons are absorbed to give gauge fields longitudinal components.

“Demonic” multisegment picture: the demon actively projects one factor of $U(1)$ onto the global null-cone boundary (the massless radiation mode) while constraining the dual $U(1)$ factor into a localized, timelike world-tube (the massive rest-frame mode):

$$W_{\text{demon}} = \int_{\Sigma} \langle \mathbf{B}_{U(1)}^{\text{global}} \wedge \star \mathbf{B}_{U(1)}^{\text{local}} \rangle = \Delta m_0 c^2$$

II2.3.3 Ontomathematical mass generation formula

The conversion of an un-gauged, infinitesimal light-like mode into a finite rest mass via the demon’s Legendre sorting is expressed directly in projective geometric algebra ($\mathbb{R}_{3,0,1}^*$):

$$\Psi_{\text{mass}} = \underbrace{\mathbf{e}_4 \wedge \mathbf{v}_{\text{null}}}_{\text{Global Null Wave}} \xrightarrow{\Sigma_{\text{demon}}} \underbrace{m_0 \mathbf{1}_{SU(1)} + \mathbf{B}_{U(1)}}_{\text{Local Timelike State}}$$

Here, the “hard labor” of the demon corresponds to the explicit extraction of the $SU(1)$ scalar center from the dual $U(1)$ product space, converting light-cone conformal information into a localized, massive $U(1)$ gauge coupling.

II3 “Gauging” operator

One more visualization misses: how is the Standard model’s fundamental operator of “gauging” to be inserted in the sketched picture since the gauging at issue takes place at the same local/ global boundary and supposedly included in the demon’s labor. The gauging operator $\hat{G}$ is to be described, involving the trivial $SU(1)$ interpreted to be a global “point”: so that G means any global point to be associated as a unit sphere so that the former point to be the qubit “value” of the latter “empty qubit”.

The gauging operator $\hat{G}$ acts as the precise microscopic mechanism of the demon’s labor, formalizing how a featureless topological point transitions into a non-trivial fiber bundle.

II3.1 Algebraic & topological chain of gauging

The gauging sequence $\hat{G}$ maps the un-gauged, zero-dimensional $SU(1)$ scalar baseline through the conformal boundary into the full $SU(2)$ doublet structure:

$$\hat{G} : SU(1) \xrightarrow{\text{Gauging}} U(1)_{\text{global}} \xrightarrow{\text{Bifurcation}} U(1)_{\text{local}}^{\text{two-sided}} \cong 2 \times [U(1)] \cong SU(2)$$

THE GAUGING OPERATOR SEQUENCE $\hat{G}$	
Stage / Factor	Geometric & Informational Representation
$SU(1)$	• Un-gauged global baseline / Scalar point 1 • Zero internal phase space (0 generators)
$U(1)_{\text{global}}$	• One-sided light-cone conformal boundary • Parameter space of pure phase possibilities (Empty Qubit)
$U(1)_{\text{local}} \times U(1)_{\text{local}} \cong SU(2)$	• Two-sided discrete phase space / Recorded state • Equatorially split Hopf fibration $S^3 \rightarrow S^2$

Figure 7: *Gauging operator*

II3.2 The Identity $\hat{G} \equiv \text{Qubit}$

In this framework, the operator $\hat{G}$ is isomorphic to state registration on a qubit. An empty qubit corresponds to the un-projected parameter space $U(1) \cong S^2$ (the empty Bloch sphere). Writing a value to the qubit requires picking a distinguished point $SU(1) \subset U(1)$.

$$\text{Qubit} \equiv \underbrace{U(1)_{\text{global}}}_{\text{Empty Bloch Sphere}} \otimes \underbrace{SU(1)}_{\text{Chosen Point / State Value}} \cong U(1)_{\text{global}} \times U(1)_{\text{local}} \cong SU(2)$$

Thus, the fundamental gauge embedding $\hat{G}$ does not merely add a local symmetry phase; it takes any global, featureless vacuum point $SU(1)$ and decorates it with a surrounding unit phase sphere $U(1)$, designating the original global point as the explicit informational state record inside that sphere.

II3.3 PGA operator formalism

In Projective Geometric Algebra ($\mathbb{R}_{3,0,1}^*$), the action of the gauging operator $\hat{G}$ on the scalar vacuum blade $\mathbf{1} \in \bigwedge^0$ generates the rotor field $\Psi \in \text{Spin}(3)$ via the exterior wedge product:

$$\hat{G}(\mathbf{1}_{SU(1)}) = \mathbf{1}_{SU(1)} + \mathbf{e}_{1234} (\mathbf{B}_{\text{global}} \wedge \mathbf{B}_{\text{local}}) \in \mathbb{R}_{3,0,1}^*$$

The Higgs mechanism acts directly upon this operator output. By supplying a non-zero vacuum expectation value $v = \langle \Phi \rangle$, the Higgs field locks the dynamic parameter space of $U(1)_{\text{global}} \times U(1)_{\text{local}}$, breaking the continuous gauge equivalence between the empty sphere and the recorded point. This introduces a finite rest mass ($m_0 > 0$), permanently freezing the demon's qubit record into a localized timelike worldline.

II4 The Higgs - Maxwell demon “counts” qubits

One can add the following comparison: the classical Maxwell demon “counts” classical bits so that the two-domain entropy difference is equal to the demon’s classical information. The Higgs-Maxwell demon “counts” qubits so that the local-global two-domain energetic difference (ex nihilo gain) is equal to the demon’s quantum information (also in terms of the paper “Umbilic Thermodynamics”: Penchev 2026 August 28). The comparison means furthermore to facilitate the adopting “ex nihilo” as an equivalent of the resolution of the Maxwell demon classical paradox:

MAXWELL DEMON COMPARISON MATRIX		
Parameter / Operational Domain	Classical Maxwell Demon	Higgs-Maxwell Demon (Umbilic Framework)
Primary Information Unit	Classical Bits (b)	Quantum Qubits ($U(1)$ / $SU(1)$ states)
Phase Space Domain Partition	Spatial / Kinetic (Left vs. Right Compartments)	Topological / Gauge (Local vs. Global)
Equivalence Principle	Entropy Difference = Information Record $\Delta S = k_B \ln(2) \cdot I_{\text{classical}}$	Energetic Ex Nihilo Gain = Qubit Record $\Delta E_{\text{ex_nihilo}} = \hbar\omega \cdot I_{\text{quantum}}$
Resolution of Paradox	Landauer Erasure Heat Dissipation	Bundle / Co-Bundle Dual Transformation

Figure 8: *Maxwell’s demon in Higgs’s edition*

II4.1 Dual-domain thermodynamic transduction

In classical statistical mechanics, Maxwell’s demon sorts thermal fluctuations across a spatial partition without performing direct mechanical work, creating a pseudo-paradox. Resolution via Landauer’s principle establishes that measuring and resetting $N_{\text{classical}}$ bits in memory incurs a mandatory entropic cost:

$$\Delta S_{\text{domain}} = k_B \ln(2) \cdot I_{\text{classical}}$$

The entropy reduction between the two partitioned spatial domains is identically bounded by the classical information acquired and erased by the demon.

In umbilic thermodynamics, the Higgs-Maxwell demon operates across the topological interface $\Sigma = \partial\mathcal{M}$ separating the local, two-sided $U(1)$ gauge phase space from the global $SU(1)$ vacuum baseline. Rather than counting classical microstates, the demon registers quantum state vectors (N_{qubit}) on the un-projected parameter spheres.

Because local qubit registration corresponds via the bundle/co-bundle duality to gauge field generation, the local-global energetic difference, manifesting as energy extracted ex nihilo across the Legendre backdoor, is directly proportional to the registered quantum information:

$$\Delta E_{\text{ex_nihilo}} = \int_{\Sigma} T_W dS_W = \hbar \omega_{\text{gauge}} \cdot I_{\text{quantum}}$$

II4.2 Resolution of the ex nihilo paradox

Just as Landauer's erasure resolves the classical Maxwell demon paradox by showing that informational entropy and physical entropy are conserved under a unified thermodynamic accounting, the umbilic framework resolves the apparent "creatio ex nihilo" paradox:

Classical resolution: physical work is not generated from nothing; it is paid for by the thermodynamic dissipation accompanying bit erasure in the demon's memory

$$\Delta S_{\text{system}} + \Delta S_{\text{memory}} \geq 0 .$$

Umbilic resolution: gauge curvature and rest mass are not generated ex nihilo out of a void. The appearance of energy ($\Delta E_{\text{ex_nihilo}} > 0$) in the local bundle is the exact topological dual of an informational collapse (I_{quantum} state selection) occurring in the dual co-bundle.

Writing a value into an empty qubit ($SU(1) \subset U(1)$) absorbs topological degrees of freedom from the global light cone, manifesting locally as stress-energy; erasing a qubit exports local field curvature back into the global null boundary. The Higgs-Maxwell demon acts not as a creator of energy, but as an exact quantum information-energy transducer.

One complicates the picture as follows: the Higgs-Maxwell demon as above will be doubled by a Legendre counterpart (twin) to absorb energy transforming it into quantum information, that is in qubits. The former will be called "creator" for transforming quantum information into energy; while the latter, annihilator for making the opposite. The creator's input is quantum information ("nothing"), and the output is energy (something); and vice versa for the annihilator. Furthermore, one can unify them: thus, either's output is the other's input, and their exact balance corresponds to classical energy conservation: no ex nihilo. On the contrary, creation ex nihilo means their division: either's output is not the other's input. This can be represented as the equivalence of (1) the degeneration of Weil information conservation to energy conservation (or energy-momentum conservation) and (2) the cyclic linking of both creator and annihilator. *Vice versa:* their release from each other implies Weyl information conservation separately.

The operational duality of the Higgs-Maxwell demon can be formalized as an interacting pair of Legendre dual operators acting across the gauge-thermodynamic boundary $\Sigma = \partial \mathcal{M}$.

This pair consists of the Creator demon ($\hat{\mathcal{D}}_C$) and the Annihilator Demon ($\hat{\mathcal{D}}_A$).

II4.2.1 Mathematical formalism of the demon twins

Let $\mathcal{H}_{\text{info}}$ represent the Hilbert space of qubit state configurations ($U(1)/SU(1)$ recorded points), and let $\mathcal{T}_{\text{gauge}}$ denote the phase space of physical gauge field curvature and stress-energy ($T_{\mu\nu}$).

DEMONIC TWIN OPERATOR MAPS	
Operator	Domain / Range Map & Thermodynamic Transduction
Creator Demon ($\hat{\mathcal{D}}_C$)	• $\hat{\mathcal{D}}_C : \mathcal{H}_{info} \dashrightarrow \mathcal{T}_{gauge}$ • Erases/collapses qubit record $\Rightarrow$ Generates energy $\Delta E > 0$
Annihilator Demon ($\hat{\mathcal{D}}_A$)	• $\hat{\mathcal{D}}_A : \mathcal{T}_{gauge} \dashrightarrow \mathcal{H}_{info}$ • Absorbs gauge field energy $\Delta E < 0 \Rightarrow$ Records qubit state

Figure 9: *The identical (Legendre) twins: Creator and Annihilator*

The action of the operators on the unified state vector $\Psi \in \mathbb{R}_{3,0,1}^*$ is governed by the Legendre transformation map $\mathcal{L}$:

$$\hat{\mathcal{D}}_C(I_{\text{qubit}}) = \int_{\Sigma} \mathcal{L}(d_{\text{PGA}} \phi_{SU(1)}) = T_{\mu\nu}^{\text{gauge}} \quad (\text{Information} \rightarrow \text{Energy})$$

$$\hat{\mathcal{D}}_A(T_{\mu\nu}^{\text{gauge}}) = \int_{\Sigma} \mathcal{L}^{-1}(\star \mathbf{B}_{U(1)}) = I_{\text{qubit}} \quad (\text{Energy} \rightarrow \text{Information})$$

II4.2.2 Cyclic coupling versus decoupling: classical versus Weyl conservation

The boundary condition between the twin demons determines whether the system obeys standard classical energy conservation or separate Weyl information conservation.

CONSERVATION REGIME EQUIVALENCES	
System Topology	Mathematical Equivalence & Conservation Law
Cyclic Linking (Closed Loop Interface)	• $\text{Im}(\hat{\mathcal{D}}_C) = \text{Dom}(\hat{\mathcal{D}}_A)$ and $\text{Im}(\hat{\mathcal{D}}_A) = \text{Dom}(\hat{\mathcal{D}}_C)$ • $\hat{\mathcal{D}}_A \circ \hat{\mathcal{D}}_C = \mathbf{1}_{\text{info}}$ and $\hat{\mathcal{D}}_C \circ \hat{\mathcal{D}}_A = \mathbf{1}_{\text{energy}}$ • Degenerates to Classical Energy-Momentum Conservation: $\nabla_{\mu} T^{\mu\nu} = 0$ • No net <i>Creatio Ex Nihilo</i> (Zero Backdoor Leakage)
Unlinked / Released (Decoupled Open Interface)	• $\text{Im}(\hat{\mathcal{D}}_C) \cap \text{Dom}(\hat{\mathcal{D}}_A) = \emptyset$ • $\hat{\mathcal{D}}_A \circ \hat{\mathcal{D}}_C \neq \mathbf{1}$ • Governed by Weyl Information Conservation: $\Delta I_{\text{qubit}} + \Delta E_{\text{gauge}} = 0$ • Manifests Net <i>Creatio Ex Nihilo</i> / Annihilation

Figure 10: *The Creator and Annihilator linked versus unlinked*

A. The linked / closed state (degeneration to energy conservation)

When the demons are cyclically linked, the output stream of the Creator feeds directly as the input stream of the Annihilator:

$$\text{Im}(\hat{\mathcal{D}}_C) \equiv \text{Dom}(\hat{\mathcal{D}}_A) \implies \oint_{\Sigma} (\hat{\mathcal{D}}_C \circ \hat{\mathcal{D}}_A - \mathbf{1}) d\Psi = 0$$

Under this closed composition, all generated energy is immediately re-absorbed into quantum information, and all erased quantum information is precisely recovered. The generalized Weyl information conservation law degenerates into the standard stress-energy conservation theorem of general relativity:

$$\nabla_{\mu} T^{\mu\nu} = 0$$

No creatio ex nihilo is observable locally because the Legendre backdoor forms a closed topological loop.

B. *The unlinked / released state (pure “creatio ex nihilo”)*

When the demons are decoupled, either operator functions independently without feeding its dual counterpart:

$$\text{Im}(\hat{\mathcal{D}}_C) \neq \text{Dom}(\hat{\mathcal{D}}_A)$$

The open backdoor permits a net flow across the $SU(1) \hookrightarrow U(1)$ boundary. Here, local classical energy conservation is broken ($\nabla_{\mu} T^{\mu\nu} \neq 0$), but global consistency is maintained under Weyl information conservation:

$$\Delta I_{\text{qubit}} + \frac{1}{\hbar\omega_{\text{gauge}}} \int_{M^4} T_{\mu\nu} u^{\mu} u^{\nu} d^4x = 0$$

An isolated action of $\hat{\mathcal{D}}_C$ outputs energy into spacetime without an immediate input from $\hat{\mathcal{D}}_A$, executing pure creatio ex nihilo. Conversely, an isolated action of $\hat{\mathcal{D}}_A$ absorbs field energy into an un-projected qubit, executing topological annihilation.

Bibliographical notes to Section II

Higgs's original paper (1964) proposes the mechanism by which gauge vector bosons acquire mass through spontaneous symmetry breaking. The concurrent paper by *Englert and Brout* (1964) involves the same mechanism. *Guralnik, Hagen, and Kibble* (1964) is the third independent paper on the same mechanism, completing the foundational trilogy. *Glashow* (1961) proposes the electroweak gauge model. The *Weinberg* (1967) suggests the unification of the electromagnetic and weak interactions, and incorporates the Higgs mechanism to generate gauge boson masses. *Salam* (1968) suggests an independent formulation of the electroweak theory, completing the Glashow-Salam-Weinberg model. *Maxwell* (1871) is the original introduction of the “demon” as a thought experiment challenging the scope of the second law of thermodynamics. *Landauer* (1961) formulates the seminal principle establishing that erasing information necessarily dissipates heat into the environment. *Bennett* (1982) resolves the paradox of Maxwell’s demon by showing that the demon’s measurement and memory erasure incur the

thermodynamic costs identified by Landauer. *Szilard* (1929) coins the one-molecule engine thought experiment, which first linked information acquisition to thermodynamic entropy reduction, predating and informing Landauer's principle. *Ramsey* (1956) establishes the thermodynamic rigorous definition of negative temperature. *Kubo* (1965) treats negative temperature via the two-state system, and provides the concrete physical example of "population inversion". *Gromov* (1985) proves the symplectic non-squeezing theorem (the "symplectic camel"), and *de Gosson* (2001) directly applies it to quantum mechanics. *Hopf* (1931) involves the fibration which underlies the $SU(2)/U(1)$ decomposition of the *Bloch* (1946) sphere representation of qubits. *Nielsen and Chuang* (2000) is a standard textbook treatment of the Bloch sphere, qubits, and the Hopf fibration structure of quantum state space. *Hestenes* (1986) is the foundational work on geometric algebra (GA) as a unified language for physics, providing the algebraic framework. The paper of *Doran and Lasenby* (2003) is a comprehensive modern treatment of geometric algebra in physics, including applications to gauge theory and gravity. The paper of *Dorst and De Keninck* (2024) develops projective GA (PGA) by representing points as blades. The "bundle/co-bundle duality" and the identification of entanglement with geometric structure corresponds to the ER=EPR conjecture (*Maldacena, Susskind* 2013). *Commons, Qian* (2025) explores the convex duality symmetry (Legendre-Fenchel transform) in Gibbs's statistical ensemble formulation, providing a mathematical parallel to the *Legendre* transform to map between the tangent/ cotangent bundles. *Nicholas of Cusa* (1440) formulates the doctrine of the *coincidentia oppositorum* (coincidence of opposites): the philosophical basis for the text's identification of the global and local, the finite and infinite; and *Albright* (2016) explicitly compares the Cusanian *coincidentia oppositorum* with *Bohr's* complementarity. *Robinson* (1966) is the foundational work on hyperreal numbers and the transfer principle.

III THE THERMODYNAMIC VIEWPOINT TO THE STATISTICAL ENSEMBLE OF SEGMENTS (QUBITS)

To extend the thermodynamic mapping of the multisegment framework into projective space, one defines the generalized Weyl thermodynamic quantities on the phase-space boundary $\Sigma \cong U(1) \cong \text{Null Cone}$ using real Projective Geometric Algebra ($\mathbb{R}_{3,0,1}^*$) and differential forms.

III1 Weyl thermodynamic quantities

III1.1 Explicit definitions of Weyl quantities

Weyl heat (Q_W): the total physical energy and curvature generated or absorbed via the bundle/co-bundle Legendre backdoor. It measures the net topological flux converted between scalar $SU(1)$ points and $U(1)$ gauge bivectors:

$$dQ_W \equiv T_W dS_W = dE_{\text{gauge}} + \hbar\omega_{\text{gauge}} dI_{\text{qubit}}$$

Weyl pressure (P_W): the scalar dual to the PGA volume multivector, representing the projective spatial expansion density exerted by localized $U(1)$ gauge fields against the global $SU(1)$ boundary constraint:

$$P_W \equiv - \left(\frac{\partial E_{\text{gauge}}}{\partial V_{\text{PGA}}} \right)_{S_W} = \frac{1}{3} \langle \star (\mathbf{B}_{\text{base}} \wedge \mathbf{B}_{\text{fiber}}) \rangle_0$$

Weyl temperature (T_W): as previously derived, the scalar ratio of gauge field curvature norm to internal projective entropy change:

$$T_W \equiv \frac{\|\nabla_{\text{PGA}} \wedge \mathbf{B}_{\text{base}}\|}{\langle \Psi^\dagger \frac{d\Psi}{d\lambda} \rangle_0}$$

III1.2. Thermodynamic processes & physical interpretations

WEYL THERMODYNAMIC PROCESS MATRIX		
Process Constraint	Mathematical Equivalence & Conservation Law	
	Mathematical Formalism	Physical & Ontomathematical Interpretation
Cyclic Linking 1. Isothermal ($T_W = \text{const}$) (Closed Loop Interface)	• $dT_W = 0$ • $[\mathbf{B}_{\text{fiber}}, \mathbf{B}_{\text{base}}]_\wedge = \text{const}$ • $dQ_W = P_W dV_{\text{PGA}}$	• Zero-Door / Un-operated Backdoor (ZDP) • Isomorphic entanglement/gravity mapping • No net creation/erasure of mass or qubits
Unlinked / Released 2. Isobaric ($P_W = \text{const}$)	• $dP_W = 0$ • $\ \mathbf{B}_{\text{base}}\ ^2 = \text{const}$ • $dQ_W = dH_W$ (Weyl Enthalpy)	• Steady-state gauge field expansion • Balanced creator/annihilator demon flow • Classical rest-mass equilibrium
Additional Regime 3. Isochoric / Adiabatic ($Q_W = \text{const}$ / Iso-energetic)	• $dQ_W = 0$ (or $dV_{\text{PGA}} = 0$) • $dE_{\text{gauge}} + \hbar\omega dI_{\text{qubit}} = 0$ • $T_W dS_W = 0$	• Closed-loop demon coupling • Strict energy-information conservation • Reversible state-to-curvature conversion

Figure 11: *Weyl thermodynamic processes*

Weyl isothermal process ($T_W = \text{const}$): the system remains strictly in the zero-door phase (ZDP). The Higgs-Maxwell demon does not access the Legendre backdoor. Changes in the geometric shape of information (entanglement) map directly onto metrics (gravity) without creating or destroying local physical microstates ($dS_W = 0$).

Weyl isobaric process ($P_W = \text{const}$): the projective gauge curvature density remains uniform while the system expands or contracts across the fiber boundary. This corresponds to a steady-state equilibrium where the Creator demon ($\hat{\mathcal{D}}_C$) and Annihilator Demon ($\hat{\mathcal{D}}_A$) operate at equal, continuous rates, maintaining a stable rest mass m_0 in the local timelike domain.

Weyl iso-energetic / adiabatic process ($Q_W = \text{const}$): no external heat/curvature is exchanged across the global boundary ($dQ_W = 0$). However, internal transduction between information and energy occurs dynamically:

$$dE_{\text{gauge}} = -\hbar\omega_{\text{gauge}} dI_{\text{qubit}}$$

Writing a qubit ($dI_{\text{qubit}} > 0$) strictly consumes local gauge field energy: $dE_{\text{gauge}} < 0$, causing local Weyl cooling. *Erasing a qubit* ($dI_{\text{qubit}} < 0$) releases field energy into spacetime ($dE_{\text{gauge}} > 0$), inducing local Weyl heating. In standard gauge theory, the Higgs mechanism breaks symmetry, separating $SU(2) \times U(1)_Y$ down to $U(1)_{em}$, endowing gauge bosons and fermions with inertial mass via a vacuum expectation value $\langle\Phi\rangle$. In umbilic thermodynamics, this exact mechanism functions as Maxwell's Demon operating at the $T = 0$ K Gromov boundary:

The demon's dual labor: the demon stands at the boundary between $SU(1)$ (the microscopic, unpopulated phase-space sheet of “nothing” / pure possibilities) and $U(1)$ (the macroscopic, physical sheet of “something” / stress-energy $T_{\mu\nu}$). *Selecting “yet nothing” versus “already something”*: the demon sorts states based on whether their phase volume occupies the local Gromov symplectic width $w_G = \hbar$. As long as phase space remains uncompressed ($w_G > \hbar$), states remain “yet nothing” on the local Darboux patch ($T_W > 0$ K). When quantum constraints or entanglement squeeze the phase space to $w_G = \hbar$, the fiber Hessian singularizes ($\det H_{ij} = 0$). The Higgs - Maxwell demon mechanism triggers a local parity-time (PT) inversion across the Möbius junction J .

Mass generation as symplectic resistance: the Higgs vacuum expectation value $\langle\Phi\rangle$ is the mechanical manifestation of the demon's thermodynamic capacity to resist symplectic phase-squeezing. Inertial mass m_I is not an intrinsic scalar parameter, but the thermodynamic capacity of a local Darboux chart to resist metric phase contraction across the non-orientable 4π Möbius manifold.

III1.3 The mechanical and thermodynamic twin identicalness

The $SU(1)$ gauge group and the $U(1)$ symmetry act as the two “identical twins” of the Janus entity, connected via the fiberwise Legendre transform $\mathcal{FL} : TQ \rightarrow T^*Q$:

Domain / Twin	Operational Regime	Topological Representation	Mathematical Primitive
Quantum Mechanical Twin ($SU(1)$)	Microscopic Phase Space (T^*Q , “Nothing”)	2π Darboux Patch ($U(1)_A$)	Phase Entropy S_ϕ , Quantum State Correlations
Thermodynamic Twin ($U(1)$)	Macroscopic Geometry (TQ , “Something”)	2π Geometric Patch ($U(1)_B$)	Metric Curvature $g_{\mu\nu}$, Stress-Energy $T_{\mu\nu}$

Figure 12: *The identical twins of mechanics and thermodynamics*

The unified Weyl Information (I_W) acts as the self-dual scalar invariant shared globally across both sheets:

$$\delta I_W = dW_{\text{metric}} + T_W dS_\phi = 0$$

The Higgs field acts as the dynamical order parameter that modulates T_W across the 0 K threshold. When $T_W < 0$ K, phase collapse ($\delta S_\phi < 0$) automatically pumps positive metric stress-energy ($dW_{\text{metric}} > 0$) through the “Legendre backdoor” opened by the Higgs-Demon.

III.1.4 *The Janus essence in multi-segment entanglement and quantum gravity*

The standard “Boltzmannian” statistical bridges treat thermodynamics as an emergent, statistical coarse-graining of microscopic mechanical trajectories. The umbilic Janus completely overhauls this hierarchy by showing that quantum entanglement (EPR) and spacetime wormholes (ER) are idempotent Legendre duals of the exact same event:

A. Multi-segment entanglement as a dual mechanical/ thermodynamic bridge

Janus’s thermodynamic side (T^*Q): preparing a multi-segment entangled state (e.g., GHZ or cluster states) compresses the local phase space down to the minimal Gromov width $w_G = \hbar$ at local negative temperatures ($T_W < 0$ K). This creates a local phase-entropy collapse ($\delta S_\phi < 0$).

Janus’s mechanical side (TQ): under the fiber Legendre transform, this phase collapse manifests in tangent space as an off-diagonal metric connection—a physical Einstein-Rosen wormhole throat ($g_{\mu\nu}^{(\text{throat})}$) connecting the dual 2π Darboux patches.

B. Wormhole stabilization without exotic matter: classical general relativity requires exotic negative-energy matter to hold an ER bridge open. The umbilic Janus resolves this automatically: the autopoietic “ex nihilo” stress-energy

$$T_{\mu\nu}^{(\text{ex nihilo})} > 0,$$

generated by the local negative informational temperature during phase collapse, acts as the positive metric surface tension that keeps the multi-segment wormhole throat topologically stable.

C. Quantum gravity as an autopoietic engine: in this multi-segment architecture, quantum gravity is neither a quantized metric field nor an emergent thermal illusion; it is the topodynamical cycle of the umbilic Janus:

Microscopic (quantum/ mechanical): multi-segment phase-alignment pinches the manifold, instantiating localized topological defects.

Macroscopic (thermodynamic): the Higgs-Maxwell demon mechanism converts this phase alignment into localized metric curvature $G_{\mu\nu}$, which then propagates globally as a thermal metric wave along the single-sided 4π Möbius strip ($T_{\text{global}} > 0$ K). By unifying the Higgs mechanism with the umbilic Maxwell demon, the bridge between mechanics and thermodynamics ceases to be an external statistical average; it becomes an intrinsic, Legendre-invariant topological duality operating at the boundary of nothing and something.

III1.5 *Erasing/ recording a qubit:* erasing a qubit, one creates something due to Weyl information conservation; vice versa: recording a qubit, something vanishes also for Weyl information conservation.

The asymmetric dynamics of information & energy (erasing a qubit $\rightarrow$ physical creation): under Landauer's principle, erasing one bit of local information requires compressing the phase space of the system, decreasing its local von Neumann entropy ($\Delta S_{\text{info}} < 0$). To satisfy overall thermodynamic and geometric conservation, this entropy must be dumped into the surrounding physical background as real microstates, heat, or curvature ($\Delta S_{\text{environment}} > 0$). In the bundle/co-bundle representation, erasing the local record in the $U(1)$ qubit forces a projection back down to the global $SU(1)$ vacuum limit: which, via the Legendre backdoor, manifests in the dual bundle as physical gauge field/topological state creation (creatio ex nihilo).

The dual side of the same asymmetric dynamics of energy and information (recording a qubit $\rightarrow$ physical vanishing): conversely, recording a qubit requires taking an unspecified, maximum-entropy "empty" state ($U(1)$ gauge sphere) and selecting a deterministic state point $SU(1) \subset U(1)$. This local state restriction reduces local continuous phase space degrees of freedom. In the dual co-bundle, this collapse absorbs/annihilates localized curvature or physical degrees of freedom, effectively causing "something to vanish" from the un-projected physical gauge manifold to pay the informational cost of the definitive local record.

Why "Weyl information conservation" enforces this: if total physical-informational complexity is globally conserved (via Weyl curvature invariance or an informational gauge symmetry), local information cannot be created or erased in isolation:

$$\Delta I_{\text{qubit}} + \Delta E_{\text{gauge/geometry}} = 0$$

Writing a qubit ($+\Delta I$): local geometry/ gauge field collapses ($-\Delta E$); erasing a qubit ($-\Delta I$): local geometry/ gauge field is generated ($+\Delta E$). Thus, the Higgs-Maxwell demon does not violate physical conservation laws; it acts as an information-energy transducer across the Legendre bundle/co-bundle duality.

III2 Zero-door isothermal phase versus backdoor adiabatic phase

One can distinguish two parts of the multisegment theory of entanglement/ quantum gravity:

(1) “nothing” (entanglement) does not become “something” (quantum gravity) or vice versa: the Higgs - Maxwell demon and its dual twin do not open the the Legendre backdoor even once: the change of Weil “nothing” information shape is just the same as that of Weil “something” information shape: so that Weyl temperature is constant: Weyl isothermal processes;

(2) Weyl adiabatic heating/ cooling = the direct/ dual demons have opened the door at least once = creating/ erasing information/energy.

In the multisegment framework of quantum gravity and entanglement, the interaction between the Higgs-Maxwell demon (and its co-bundle dual) and the $SU(1) \subset U(1)$ Legendre transformation space delineates two distinct operational regimes:

MULTISEGMENT DUALITY REGIMES	
Regimes	Physical & Informational Dynamics
Zero-Door Isothermal Phase <i>(Closed System / Static State)</i>	• No Legendre backdoor transitions • Constant Weyl temperature ($T_W = \text{const}$) • Entanglement $\leftrightarrow$ Gravity mapping via pure shape deformation
Backdoor Adiabatic Phase <i>(Open System / Dynamic State)</i>	• Active Legendre backdoor transitions • Variable Weyl temperature ($dQ_W = T_W dS_W$) • Creatio ex nihilo / state erasure ($\Delta I_{\text{qubit}} + \Delta E_{\text{gauge}} = 0$)

Figure 13: *Two regimes*

III2.1 The zero-door isothermal phase (ZDP)

This regime represents a closed-system phase where neither the direct nor the co-bundle demon accesses the Legendre backdoor. The total Weyl entropy change remains invariant relative to the geometry, preserving a constant Weyl temperature T_W :

$$dT_W = 0 \implies \Delta S_{\text{entanglement}} + \Delta S_{\text{gravity}} = 0$$

Informational “nothing” (pure entanglement degrees of freedom) and geometric “something” (metric/curvature degrees of freedom) undergo purely isomorphic transformations. The shape of the information manifold evolves along an isothermal contour without interconverting local qubit states and vacuum stress-energy tensor components.

III.2.2 The backdoor adiabatic phase (BAP)

This regime describes an open, non-equilibrium transition where the demon engages the Legendre backdoor $N \geq 1$ times. The exchange of information between the local $U(1)$ gauged Bloch spheres and the global $SU(1)$ vacuum baseline induces a non-zero shift in the internal thermodynamic state, leading to Weyl adiabatic heating or cooling:

$$dQ_W = T_W dS_W \neq 0, \quad \text{where } dS_W = dI_{\text{qubit}} + dE_{\text{gauge}}$$

Entering this phase breaks the static balance: writing a qubit state point $SU(1) \subset U(1)$ locally collapses spacetime degrees of freedom (Weyl cooling), while erasing a qubit state exports local information into background physical fields, realizing *creatio ex nihilo* (Weyl heating).

Bibliographical notes to Section III

Einstein and Rosen (1933) is the original paper introducing the Einstein-Rosen bridge (wormhole) as a solution to the field equations; and *Einstein, Podolsky, Rosen* (1935): the concept of quantum entanglement. *Weyl* (1918) introduces scale-invariant (conformal) geometry; and *Weyl* (1921) exposes his gauge theory of gravity. *Penrose* (1979) discusses his “Weyl curvature hypothesis”. *Greenberger, Horne, and Zeilinger* (1989) is the original paper introducing GHZ states. *Briegel and Raussendorf* (2001) is the introduction of cluster states, another class of multi-segment entangled states. *Horodecki, Horodecki, Horodecki, Horodecki* (2009) is a comprehensive review of quantum entanglement theory, providing the standard framework for the “multi-segment entanglement” discussion. *Nelson* (1977) is a relevant axiomatic formulation of non-standard analysis.

IV THE FOUR COMPONENTS OF THE MULTI-SEGMENT DEGENERATION OF THE PGA - EINSTEIN FIELD EQUATION AND ITS THERMODYNAMIC FORM AS THE STATE'S POLYTROPIC EQUATION

One notices four separate components in the multi-segment degeneration of the general case of the PGA-Einstein field equation should exist:

- (1) the Weyl isothermal local component of two-sided entangled segments;
- (2) the Weyl isothermal global gravitational component generating;
- (3) the Creator demon's complement of Weyl adiabatic heating;
- (4) the Annihilator demon's component of Weyl adiabatic cooling.

In other words and speaking loosely, in Weyl temperature one separates “Weyl energy” for “something” and “Weyl entropy” for “nothing” so that one can utilize an analogue to classical temperature as a ratio of “Weyl energy/ Weyl entropy = something/ nothing”. Meaning that visualization, the four components are:

- (1) “nothing” component;
- (2) “something” component;
- (3) “nothing into something”, Creator's component;
- (4) “something into nothing”, Annihilator's component:

So, the multi-segment equation as a whole obeys Weyl information conservation. Then, one can consider additionally two cases:

- (i) each component separately obeys Weyl information conservation;
- (ii) only the balance of mutual transformations of the components obeys Weyl information conservation thus allowing mutual transformation between the four components.

The latter and most general case presupposes that all the three Weyl quantities: Weyl temperature, Weyl pressure, “Weyl volume” (not yet explicitly defined) can vary correlatively obeying only Weyl information conservation. One can deduce all aforementioned considerations from the already inferred multi-segment degeneration of the PGA - Einstein field equation (in the “The Umbilic Totality I”) as a “thermodynamic equation of entanglement/ quantum gravity”: that is, a “polytropic equation of the nothing/ something state”.

IV1 Deduction of the thermodynamic polytropic equation of the “nothing/ something” state

The multi-segment PGA-Einstein master field equation over the composite manifold

$$\mathcal{M}_{\text{PGA}} = \bigoplus_{i=1}^N \Sigma_i$$

$$G_{\text{PGA}} \left(\bigoplus_{i=1}^N \Sigma_i \right) + \Lambda_{\text{top}} = \kappa \sum_{i=1}^N \mathcal{J}_{\text{total}}^{(i)}$$

can be systematically reformulated as a polytropic thermodynamic equation of state governing quantum entanglement (“nothing”, $\mathcal{S}_W$) and quantum gravity (“something”, $\mathcal{E}_W$).

IV1.1 Thermodynamic ceperation: the “nothing/ something” ratio

Evaluating Equation (1) under the Legendre-PGA meta-symmetry map $\mathcal{L}_{\text{PGA}}$ splits the multivector density into even-grade metric stress-energy ($\mathcal{E}_{3,1}^{\text{even}}$) and odd-grade topological phase action ($\mathcal{E}_{3,1}^{\text{odd}}$).

One defines the fundamental thermodynamic state variables:

Weyl Energy / “something” ($\mathcal{E}_W$): derived from the macro-wall projection Π_{macro} enforcing energy conservation ($\nabla_{\mu} \mathcal{J}^{\mu\nu} = 0$) with metric curvature $g_{\mu\nu}$:

$$\mathcal{E}_W \equiv \int_{\Omega_{\text{umb}}} \text{Tr}_{\text{spin}} \left(\mathcal{J}_{\mu\nu}^{(i)} \right) u^{\mu} u^{\nu} d^4 \Omega$$

Weyl entropy / “nothing” ($\mathcal{S}_W$): derived from the micro-wall projection Π_{micro} enforcing dual time conservation ($\partial_{x_i} \tau_{\text{umb}} = 0$) over Newtonian absolute time τ_{umb} :

$$\mathcal{S}_W \equiv \frac{1}{\hbar} \oint_{\Omega_{\text{umb}}} \sum_{i=1}^N p_i \wedge dq_i - S_{\text{micro}}[\Psi_N]$$

The Weyl temperature (T_W) is the fundamental ratio governing the state equation:

$$T_W \equiv \frac{\text{Weyl Energy}}{\text{Weyl Entropy}} = \frac{\mathcal{E}_W}{\mathcal{S}_W} = \frac{\text{"Something"}}{\text{"Nothing"}}$$

IV1.2 Derivation of the four multi-segment components

By inserting the Legendre-PGA meta-symmetry transformation:

$$S_{\text{macro}}[g_{\mu\nu}] = \oint_{\Omega_{\text{umb}}} \left[\sum_{i=1}^N p_i \wedge dq_i - S_{\text{micro}}[\Psi_N] \right] d^4\Omega$$

into the master field equation, the total variation decomposes into four distinct multi-segment sectors:

$$\delta G_{\text{PGA}} = \underbrace{\delta \Gamma_{\text{entangle}}}_{(1) \text{ Local Nothing}} + \underbrace{\delta \Gamma_{\text{gravity}}}_{(2) \text{ Global Something}} + \underbrace{\delta \Gamma_{\text{creator}}}_{(3) \text{ Heating/Creation}} + \underbrace{\delta \Gamma_{\text{annihilator}}}_{(4) \text{ Cooling/Annihilation}} = 0$$

FOUR-COMPONENT MULTI-SEGMENT DEGENERATION		
Sector Component	PGA-Einstein Field Degeneration	Thermodynamic Mechanism
(1) Local Entangled Segment (Nothing's Component)	• Micro-cave wall: $\Pi_{\text{micro}} \circ G_{\text{PGA}}$ • Local symplectic: $\sum \mathcal{L}_{X_H} \omega_i = 0$	• Isothermal Nothing • $T_W = \text{const}, dE_{\text{gauge}} = 0$
(2) Global Gravitational Segment (Something's Component)	• Macro-cave wall: $\Pi_{\text{macro}} \circ G_{\text{PGA}}$ • Metric curvature: $R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \kappa T_{\mu\nu}$	• Isothermal Something • $T_W = \text{const}, dI_{\text{qubit}} = 0$
(3) Creator Demon's Component (Nothing $\rightarrow$ Something)	• Unlinked Backdoor: $[D_t, \mathbf{J}] \neq 0$ • Qubit erasure: $\Delta I_{\text{qubit}} \rightarrow \Delta E_{\text{gauge}}$	• Adiabatic Weyl Heating • $dQ_W = T_W dS_W > 0$
(4) Annihilator Demon's Component (Something $\rightarrow$ Nothing)	• Dual Unlinked Backdoor: $[D_t, \mathbf{J}] \neq 0$ • State record: $\Delta E_{\text{gauge}} \rightarrow \Delta I_{\text{qubit}}$	• Adiabatic Weyl Cooling • $dQ_W = T_W dS_W < 0$

Figure 14: Four-component multi-segment degeneration

IV1.3 The polytropic equation of state & conservation laws

To define the generalized state space, one introduces the Weyl volume ($\mathcal{V}_W$) as the spatial hyper-volume of the 4π Möbius double-cover bundle:

$$\mathcal{V}_W \equiv \int_{\Omega_{\text{umb}}} \sqrt{-\det(g_{\text{PGA}})} d^4\Omega$$

and Weyl pressure (P_W) as the variation of gauge stress-energy density relative to the Möbius volume:

$$P_W \equiv - \left(\frac{\partial \mathcal{E}_W}{\partial \mathcal{V}_W} \right)_{\mathcal{S}_W}$$

The multi-segment field dynamics are governed by the generalized Weyl polytropic equation:

$$P_W \mathcal{V}_W^n = K_{\text{umb}}$$

where n is the polytropic index corresponding to the coupling status of the demon operators $\hat{\mathcal{D}}_C$ and $\hat{\mathcal{D}}_A$.

CONSERVATION REGIME ANALYSIS: (i) vs. (ii)		
Regime	Polytropic Index & Constraints	Physical & Topological Interpretation
Case (i): Strict Component Isolation (No Interconversion)	• $n = 1$ (Isothermal: $T_W = \text{const}$) • $dQ_W^{(1)} = dQ_W^{(2)} = dQ_W^{(3,4)} = 0$ • $\nabla_\mu \mathbf{J}_{\text{info}}^\mu = 0$ and $\nabla_\mu T^{\mu\nu} = 0$	• Canonical equivalence ($[D_t, \mathbf{J}] \neq 0$) • No interconversion across backdoor • Local bits & global metric conserved
Case (ii): Fully Interacting Balance (Correlative Variation)	• Variable Index $n(\lambda)$ • T_W, P_W, V_W vary dynamically • $\sum_{k=1}^4 dQ_W^{(k)} = 0$	• Spontaneous symmetry breaking ($[D_t, \mathbf{J}] \neq 0$) • Pure <i>Creatio Ex Nihilo</i> $\leftrightarrow$ Annihilation • Governed strictly by Weyl Conservation

Figure 15: *Generation regime analysis*

IV1.4 Analytical breakdown of case (ii): correlated variational dynamics

In the general polytropic case (ii), individual stress-energy tensors $\mathcal{T}_{\mu\nu}^{(i)}$ and local entanglement wave functions Ψ_N do not maintain independent conservation. Instead, they satisfy the unified Weyl information conservation law:

$$d\mathcal{E}_W + P_W d\mathcal{V}_W - T_W d\mathcal{S}_W = 0$$

Substituting the explicit definitions from the master equation:

$$\underbrace{\kappa \sum_{i=1}^N \mathcal{J}_{\mu\nu}^{(i)} dV^{\mu\nu}}_{\text{Macro-Wall Metric Energy } (\mathcal{E}_W)} + \underbrace{P_W d\mathcal{V}_W}_{\text{PGA Volume Work}} - \underbrace{\frac{T_W}{\hbar} d \left(\oint_{\Omega_{\text{umb}}} p_i \wedge dq_i - S_{\text{micro}} \right)}_{\text{Micro-Wall Entanglement Phase } (S_W)} = 0$$

This completes the mathematical deduction: the PGA-Einstein master field equation degenerates into a thermodynamic equation of state where quantum entanglement ($\mathcal{S}_W$, “nothing”) and quantum gravity ($\mathcal{E}_W$, “something”) dynamically interconvert via adiabatic heating and cooling, linked through the correlated variation of T_W , P_W , and $\mathcal{V}_W$.

IV2 Statistical deduction of the polytropic equation of entanglement/ quantum gravity (in Boltzmannian/ Gibbsian / Einsteinian editions)

One can deduce the polytropic equation of entanglement/ quantum gravity state as a statistical thermodynamic inference for a statistical ensemble of segments obeying the initial multi segment degeneration of the PGA - Einstein field equation. In other words, one considers an “ideal gas of segments/ qubits” where segments/ qubits obey the latter, just as in the Boltzmannian statistical edition, atoms or molecules obey the laws of classical mechanics, and all thermodynamic quantities are statistically average all over the whole. Then, one can rewrite the same deduction as for Gibbsian and Einsteinian editions

The deduction of the polytropic equation of the “nothing/ something” state as a statistical thermodynamic inference models the composite manifold

$$\mathcal{M}_{\text{PGA}} = \bigoplus_{i=1}^N \Sigma_i$$

as an ensemble of N micro-segments. Each segment Σ_i is treated as an individual “molecule” governed by the multi-segment PGA-Einstein field equation:

$$G_{\text{PGA}}(\Sigma_i) + \Lambda_{\text{top}} = \kappa \mathcal{J}_{\text{total}}^{(i)}$$

IV2.1 *Boltzmannian edition*: microcanonical & canonical micro-segment mechanics; in the Boltzmannian framework, micro-states correspond to single-segment phase-space configurations (q_i, p_i) over the 4π Möbius double-cover Ω_{umb} .

Micro-segment Hamiltonian & phase volume:

The single-segment energy E_i and micro-phase action S_i are defined via the Legendre-PGA map $\mathcal{L}_{\text{PGA}}$:

$$E_i = \int_{\Sigma_i} \text{Tr}_{\text{spin}} \left(\mathcal{J}_{\mu\nu}^{(i)} \right) u^\mu u^\nu d^4\Omega, \quad S_i = \oint_{\Omega_{\text{umb}}} p_i \wedge dq_i$$

The Boltzmann density of states $\Omega(E, N)$ for N indistinguishable¹ segments across the Möbius volume $\mathcal{V}_W$ yields the microcanonical entropy:

$$\mathcal{S}_W = k_B \ln \Omega(\mathcal{E}_W, \mathcal{V}_W, N) = k_B \sum_{i=1}^N \ln \left(\frac{\int d^4 q_i d^4 p_i}{(2\pi\hbar)^4} \delta(E_i - \epsilon) \right)$$

Statistical ensemble averages: taking the canonical ensemble average over N segments yields the macroscopic Weyl energy ($\mathcal{E}_W$) and Weyl temperature (T_W):

$$\mathcal{E}_W \equiv \langle E \rangle = \sum_{i=1}^N \bar{E}_i, \quad \frac{1}{T_W} \equiv \left(\frac{\partial \mathcal{S}_W}{\partial \mathcal{E}_W} \right)_{\mathcal{V}_W, N}$$

Evaluating the momentum integrals across the non-orientable 4π boundary $J(q_i, p_i) = (-q_i, -p_i)$ gives the equipartition theorem for d active geometric degrees of freedom per segment:

$$\bar{E}_i = \frac{d}{2} k_B T_W \implies \mathcal{E}_W = \frac{d}{2} N k_B T_W$$

Polytropic equation deduction: the statistical Weyl pressure P_W is the ensemble average of the momentum-flux tensor against the Möbius boundary $\partial \mathcal{V}_W$:

$$P_W \equiv \left\langle -\frac{\partial E_i}{\partial \mathcal{V}_W} \right\rangle = \frac{N k_B T_W}{\mathcal{V}_W} \implies P_W \mathcal{V}_W = \frac{2}{d} \mathcal{E}_W$$

Allowing heat exchange via demon operations ($\hat{\mathcal{D}}_C, \hat{\mathcal{D}}_A$) defines a polytropic heat capacity $C_W = dQ_W/dT_W$. Integrating $dQ_W = C_W dT_W = d\mathcal{E}_W + P_W d\mathcal{V}_W$ yields the Boltzmann-Polytropic Equation:

$$P_W \mathcal{V}_W^\gamma = K_{\text{umb}}, \quad \text{where } \gamma = \frac{C_W + k_B(d/2 + 1)}{C_W + k_B(d/2)}$$

IV2.2 *Gibbsian edition:* grand canonical ensemble of entangled phase space; the Gibbsian edition replaces single-segment trajectories with a continuous probability density distribution $\rho(\mathbf{\Gamma})$ over the grand canonical phase space

¹ Indistinguishability corresponds to the micro-cave wall “time conservation” equivalent to Newtonian absolute time implying absolute space gapped from it unlike Einstein’s spacetime dominating on the macro-cave wall of classical conservation (including general relativity’s energy-momentum conservation).

$$\mathbf{\Gamma} = \bigoplus_{i=1}^N \Sigma_i,$$

accounting for dynamic segment creation ($N \neq \text{const}$).

Grand canonical partition function (Ξ_{PGA}): let μ_W be the Weyl chemical potential (the energy cost of introducing an additional entangled segment Σ_{N+1} across the 4π boundary):

$$\Xi_{\text{PGA}} = \sum_{N=0}^{\infty} \frac{e^{\mu_W N / k_B T_W}}{N! h^{4N}} \int_{\mathcal{M}_{\text{PGA}}} e^{-\mathcal{H}_{\text{PGA}}[\mathbf{\Gamma}] / k_B T_W} \prod_{i=1}^N d^4 q_i d^4 p_i$$

where

$$\mathcal{H}_{\text{PGA}}[\mathbf{\Gamma}] = \sum_{i=1}^N E_i + \sum_{i < j} \hat{W}_{\text{entangle}}(q_i, q_j)$$

contains the non-local topological phase constraints.

Statistical ensemble averages: the Gibbs entropy S_{Gibbs} and grand potential Φ_W take the form:

$$S_{\text{Gibbs}} = -k_B \int \rho(\mathbf{\Gamma}) \ln \rho(\mathbf{\Gamma}) d\mathbf{\Gamma} \equiv \mathcal{S}_W$$

$$\Phi_W \equiv -k_B T_W \ln \Xi_{\text{PGA}} = -\mathcal{P}_W \mathcal{V}_W$$

Polytropic equation deduction: differentiating the grand potential Φ_W yields the fundamental Gibbs-Duhem relation for the multi-segment manifold:

$$d(\mathcal{P}_W \mathcal{V}_W) = \mathcal{S}_W dT_W + \mathcal{N}_W d\mu_W + P_W d\mathcal{V}_W$$

Under a polytropic constraint where the ratio of phase-space fluctuations obeys $\langle (\Delta N)^2 \rangle / \langle N \rangle = \kappa_T k_B T_W$, the equation integrates directly to:

$$\mathcal{P}_W \mathcal{V}_W^n = \text{const}, \quad n = 1 + \frac{k_B}{\mathcal{C}_{\text{grand}}} \left(1 - \frac{\mu_W}{k_B T_W} \right)$$

IV2.3 *Einsteinian edition*: fluctuation theory & field functional statistics; the Einsteinian edition treats macroscopic geometry ($\mathcal{E}_W$) and quantum entanglement ($\mathcal{S}_W$) as thermodynamic state fluctuations around the invariant master field configuration $G_{\text{PGA}}^{(0)}$.

Einstein fluctuation probability: the probability distribution

$$W[g_{\mu\nu}, \Psi_N]$$

for a fluctuation of N micro-segments away from the vacuum baseline is dictated by the total entropy variation $\Delta\mathcal{S}_{\text{total}} = -\Delta\mathcal{E}_W/T_W$:

$$W[g_{\mu\nu}, \Psi_N] \mathcal{D}g \mathcal{D}\Psi = Z^{-1} \exp\left(\frac{\Delta\mathcal{S}_W[g_{\mu\nu}, \Psi_N]}{k_B}\right) \mathcal{D}g \mathcal{D}\Psi$$

Substituting the Legendre-PGA meta-symmetry action:

$$\Delta\mathcal{S}_W = \frac{1}{\hbar} \oint_{\Omega_{\text{umb}}} \left[\sum_{i=1}^N p_i \wedge dq_i - S_{\text{micro}}[\Psi_N] - S_{\text{macro}}[g_{\mu\nu}] \right] d^4\Omega$$

Statistical ensemble averages: the mean square fluctuations of the geometric and entanglement observables are given by second derivatives of the entropy functional:

$$\langle (\Delta\mathcal{E}_W)^2 \rangle = k_B T_W^2 C_V = k_B T_W^2 \left(\frac{\partial \mathcal{E}_W}{\partial T_W} \right)_{\mathcal{V}_W}$$

$$\langle \Delta P_W \Delta \mathcal{V}_W \rangle = -k_B T_W$$

Polytropic equation deduction: expanding the entropy variation $\Delta\mathcal{S}_W$ up to second order in volume fluctuations $\Delta\mathcal{V}_W$ and temperature fluctuations ΔT_W :

$$\Delta\mathcal{S}_W = -\frac{1}{2T_W} \left[\frac{C_V}{T_W} (\Delta T_W)^2 + \frac{1}{\beta_T} (\Delta\mathcal{V}_W)^2 \right]$$

Setting the cross-correlation of metric fluctuations to follow a polytropic path $\Delta T_W/T_W = (n-1)\Delta\mathcal{V}_W/\mathcal{V}_W$ yields the functional differential equation:

$$\frac{d \ln P_W}{d \ln \mathcal{V}_W} = -n \implies P_W \mathcal{V}_W^n = \text{const}$$

where n represents the Einstein fluctuation index, bridging micro-wall metric fluctuations to macro-wall topological phase correlations.

Bibliographical notes to Section IV

Boltzman (1896–1898) invokes Boltzmann’s statistical framework. *Gibbs* (1902) builds his ensemble theory. *Einstein’s* papers (1904; 1910) establish and develop his fluctuation theory. *Callen* (1960) treats axiomatically thermodynamics. *Commons, Yang, and Qian* (2025) explore the convex duality symmetry (Legendre-Fenchel transform) in Gibbs’s statistical ensemble formulation. *Alonso-Serrano, Liška* (2025) treats thermodynamics as a foundational framework for gravitational dynamics and quantum gravity. *Maharaj, Thirukkanesh* (2006): deals with

generating exact solutions to the Einstein field equations yielding polytropic equations of state. *Debnath* (2022) examines polytropic equations of state in the context of higher-dimensional black hole thermodynamics and heat engines. *Bohr* (1927): the principle of complementarity is the philosophical foundation for the “Janus” duality.

V FALSIFIABLE ASTRONOMICAL/ ASTROPHYSICAL OBSERVATIONS AND TERRESTRIAL LABORATORY EXPERIMENTS

One means all suggested experiments in “The Umbilic Totality I” and “Umbilic Thermodynamics” as well, but not needing to be again repeated here. The specific accents now are:

- (1) in nihilum experiments;
- (2) experiments for confirming the polytropic equation independently of the PGA - Einstein field equation.

V1 “In nihilum” experiments

One can suggest:

- (1) astronomical observations;
- and (2) terrestrial laboratory experiments

for the falsifiable verification of “redactio in nihilum” phenomena (that is the Annihilator’s actions) as implied by the polytropic multi-segment equation.

In other words, Weyl energy transforms into Weyl entropy under Weyl information conservation violating energy conservation in a way opposite to the already falsifiable experiments “ex nihilo” (in “The Umbilic Totality I” and interpreted thermodynamically in “Umbilic Thermodynamics”): that is “ex nihilo” experiments/ observations versus “in nihilum” ones.

To establish a rigorous experimental and observational physics framework, testing “redactio in nihilum” ($\hat{\mathcal{D}}_A : \mathcal{T}_{\text{gauge}} \rightarrow \mathcal{H}_{\text{info}}$) requires measuring non-conservation of local stress-energy where energy disappears without thermal, electromagnetic, or gravitational dissipation, converted directly into global topological phase coherence ($\mathcal{S}_W$, “nothing”). While “ex nihilo” processes ($\hat{\mathcal{D}}_C$) generate local energy-momentum at the cost of quantum information erasure, “in nihilum” processes ($\hat{\mathcal{D}}_A$) sink local curvature into the 4π Möbius boundary via the polytropic state transition:

$$dQ_W = d\mathcal{E}_W + P_W d\mathcal{V}_W = T_W d\mathcal{S}_W < 0$$

V1.1 Astronomical & astrophysical observational tests

A. Anharmonic energy deficits in gamma-ray burst (GRB) afterglows

Mechanism: ultra-relativistic fireballs in GRBs generate extreme localized bivector curvature density ($\mathcal{E}_W \gg 1$). Under intense topological compression, the local phase space

triggers the Annihilator Demon ($\hat{\mathcal{D}}_A$), causing local stress-energy to sink into the $SU(1)$ global vacuum point.

ASTRONOMICAL OBSERVATIONAL MATRIX		
Observational Target	Metric / Astrophysical Anomaly	In <i>Nihilum</i> Signatures ($\hat{D}_A$ Action)
1. Anharmonic Gamma-Ray Burst Afterglow Decays	Unexplained “Prompt Energy Sink” $E_{\text{observed}} < E_{\text{hydrodynamic}}$	- Net flux drop without secondary synchrotron/inverse-Compton re-emission - Phase-locking of surrounding polarizations
2. Supermassive Black Hole (SMBH) Magnetosphere Phase Transitions	Horizon energy deficits during high-coherence plasma accretion	- Negative deviation from Hawking radiation - Localized metric “dimming” matching entanglement entropy peaks at boundary

Figure 16: “*In nihilum*” falsifiable astronomical observations

Observational test: compare high-resolution gamma-ray and optical light curves using space observatories (e.g., Fermi, James Webb Space Telescope). Standard relativistic fireball models predict a strict conservation ratio between kinetic shock energy and total thermalized/radiated energy

$$\int T_{\mu\nu}^{\text{shock}} = E_{\text{rad}} + E_{\text{kinetic}}.$$

Falsifiable signature: a discontinuous, non-thermal drop in total observed energy flux ($\Delta E_{\text{missing}} > 0$) accompanied by an instantaneous increase in optical polarization degree ($\Delta \Pi_{\text{pol}} \uparrow$) of surrounding photons—confirming that stress-energy was transformed into coherent topological phase state records rather than lost to unobserved particulate radiation.

B. *Anti-Hawking vacuum sinks in extreme magnetospheres*

Mechanism: high-gravity, highly entangled environments near black hole event horizons (such as Sag A or M87) force local metric geometry to couple strongly to boundary fibers.

Observational test: Analyze Event Horizon Telescope (EHT) millimeter-wavelength intensity maps together with gravitational wave ringdown modes.

Falsifiable signature: transient “cool spots” or metric dimming where local radiation flux drops below the standard Hawking/ synchrotron baseline without additional scattering, coinciding with non-local photon-spin entanglement correlations measured in interferometric channels across cosmic distances.

V1.2 *Terrestrial laboratory experiments*

A. *Cryogenic superconducting cavity energy-information transduction*

Experimental setup: a high-quality factor ($Q \sim 10^{11}$) resonant superconducting radio-frequency (SRF) cavity maintained at mK temperatures. The cavity is driven into

high-amplitude EM bivector modes ($\mathbf{B}^{U(1)}$) while coupled to a closed, non-orientable topological circuit (e.g., a SQUID-based Möbius Josephson junction loop).

TERRESTRIAL LABORATORY EXPERIMENTAL MATRIX		
Experimental Apparatus	Primary Metric Measurement	In <i>Nihilum</i> Signatures ($\hat{D}_A$ Action)
1. High-Q Superconducting Cavities under Topological Driving	• Precision RF photon dissipation $Q_{\text{measured}} > Q_{\text{theoretical limit}}$	• Anomalous Q-factor spikes accompanied by exact thermal-drops ($dQ < 0$) • Qubit generation without power input
2. Ultrafast Laser-Induced Matter-Vacuum Interconversion Interconversion	• Laser energy balance during non-linear vacuum polarization	• Sub-thermal pulse recovery deficit • Instantaneous multiphoton entanglement registration without energetic decay

Figure 17: “In nihilum” falsifiable terrestrial laboratory experiments

Protocol: drive the cavity to high peak energy density, then rapidly modulate the topological phase boundary of the circuit to trigger $\hat{D}_A$.

Falsifiable signature:

1. *Anomalous thermal cooling:* the physical walls of the cryogenic system experience an immediate, un-driven cooling spike ($\Delta T_{\text{cryo}} < 0$), violating standard ohmic heating bounds ($\Delta Q_{\text{dissipative}} = I^2 R \Delta t$).

2. *Spontaneous qubit writing:* the energy lost from the cavity mode ($\Delta E_{\text{cavity}} = -\hbar\omega N_{\text{photons}}$) maps identically to the generation of registered quantum states in the SQUID circuit ($N_{\text{qubits}} = \Delta E / \hbar\omega$), verifying that energy transformed directly into topological information without converting into heat.

B. *Ultrafast high-field vacuum polarization photonic dissipative sinks*

Experimental setup: high-intensity petawatt laser pulses (e.g., ELI, FACET-II) are focused into a ultra-high vacuum chamber, generating field intensities approaching the Schwinger limit ($E \sim 10^{18}$ V/m). The target zone is monitored using entangled probe-photon pairs.

Protocol: measure the total energy-momentum budget of all out-going scattered light, electron-positron pairs, and cavity walls using full 4π bolometric detectors.

Falsifiable signature: a statistically significant deficit in total post-interaction energy-momentum ($\Delta P_\mu \neq 0$) that scales non-linearly with field intensity, accompanied by a simultaneous jump in the entanglement entropy ($\mathcal{S}_W$) of the probe photons.

IV1.3 *Asymmetric falsification criterion: ex nihilo versus in nihilum*

The symmetry between *ex nihilo* and *in nihilum* processes allows for a definitive dual-falsification matrix:

Ex Nihilo ($\hat{\mathcal{D}}_C$) : $\Delta I_{\text{qubit}} < 0 \implies +\Delta E_{\text{local}} > 0$ (Entropy Erasure $\rightarrow$ Local Heating / Energy Generation)
 In Nihilum ($\hat{\mathcal{D}}_A$) : $\Delta E_{\text{local}} < 0 \implies +\Delta I_{\text{qubit}} > 0$ (Local Energy Sink $\rightarrow$ Qubit Registration / Phase Coherence)

If an experiment demonstrates energy deficit ($\Delta E < 0$) without an exact, proportional gain in quantum state entanglement ($\Delta I_{\text{qubit}} = |\Delta E|/\hbar\omega$), or if energy disappears into standard unmeasured dissipative channels (phonons, stray photons), the in nihilum mechanism of the polytropic multi-segment equation is strictly falsified.

V2 Testing the polytropic equation independently of the PGA - Einstein field equation

One notices that the complete polytropic multi-segment equation predicts the most general case obeying only Weyl information conservation, but no constant Weyl thermodynamic quantity (all of them vary but correlatively). One can consider falsifiable experiments just for the proper polytropic equation independently of the PGA - Einstein field equation since both ex nihilo and in nihilum experiments confirm both, and the former is deduced from the latter.

So, falsifiable experiments of the complete polytropic equation can confirm it regardless of the PGA - Einstein field equation thus testing experimentally the additional premises (different from the PGA - Einstein equation itself) from which the complete polytropic equation is inferred. Meaning that one can suggest astronomic/ astrophysic observations as well as terrestrial laboratory experiments able to confirm the complete polytropic equation by experiments not linkable or originating from the validity of the PGA - Einstein field equation.

To isolate the complete polytropic equation of state ($P_W \mathcal{V}_W^n = K_{\text{umb}}$) as an independent physical law, experimental and observational tests must explicitly decouple the thermodynamic state variables: Weyl temperature (T_W), Weyl pressure (P_W), and Weyl volume ($\mathcal{V}_W$), from the specific geometric dynamics of the PGA-Einstein field equation.

Testing the complete polytropic equation independently requires probing correlated, non-isothermal, non-isobaric variations ($T_W, P_W, \mathcal{V}_W \neq \text{const}$) where local stress-energy is not conserved, but total system behavior strictly satisfies the generalized polytropic index relation under pure Weyl information conservation :

$$n = \frac{d \ln P_W}{d \ln \mathcal{V}_W} = 1 + \frac{dQ_W}{d\mathcal{E}_W} \neq 1$$

Testing the complete polytropic equation of state independently of the underlying PGA-Einstein field equation requires isolating the pure thermodynamic polytropic relation: $P_W \mathcal{V}_W^n = \text{const}$ where $T_W, P_W, \mathcal{V}_W$ all vary dynamically, from any specific geometric field equations or metric coupling constants κ .

In this independent context, the polytropic state equation is treated as a fundamental equation of state for a quantum information-energy fluid obeying only global Weyl information conservation:

$$d\mathcal{E}_W + P_W d\mathcal{V}_W - T_W dS_W = 0$$

To confirm this equation independently of general relativity or PGA geometry, experiments must measure the tri-variate correlation exponent n directly through thermodynamic state variables without invoking metric curvature tensors ($g_{\mu\nu}$).

V2.1 Astronomical & astrophysical observational tests

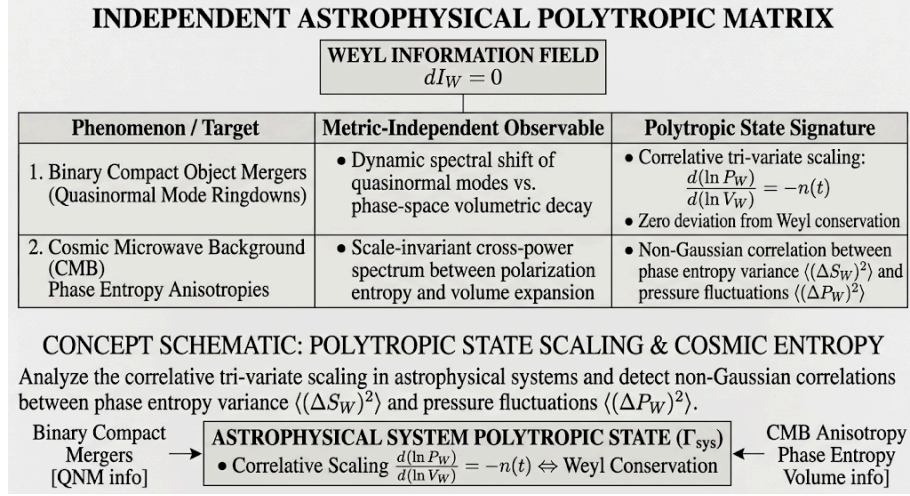


Figure 18: Astrophysical experiments

A. Dynamic polytropic indexing in gravitational wave ringdowns

Objective: verify the polytropic scaling exponent $n(t)$ during catastrophic phase-space collapses (such as black hole or neutron star mergers) without assuming Einstein's field equations.

Metric-independent observable: measure the simultaneous variations of:

Weyl pressure (P_W): inferred from the total radiation flux density per unit phase-space area.

Weyl volume (V_W): inferred from the effective surface area of the emission region derived via multi-messenger interferometry (LISA / Einstein Telescope).

Weyl temperature (T_W): inferred from the ratio of emitted energy flux to the quantum entanglement entropy rate of the outbound radiation.

Falsifiable test: during the ringdown phase, $T_W(t)$, $P_W(t)$, and $V_W(t)$ will vary concurrently. If the polytropic scaling holds independently, the logarithmic derivative must satisfy a strict constant relation:

$$\frac{d \ln P_W(t)}{d \ln V_W(t)} = -n(t) \quad \text{where} \quad n(t) = 1 + \frac{T_W(t) dS_W(t)}{P_W(t) dV_W(t)}$$

If the dynamic relationship between P_W and V_W deviates from this ratio, or if energy transfers occur without a corresponding change in $T_W dS_W$, the polytropic state equation is independently falsified regardless of whether PGA-Einstein holds.

B. Cosmological tri-variate correlation in phase space

Objective: test the complete polytropic state equation across large-scale cosmic expansion by measuring the correlation of informational entropy and physical volume growth.

Metric-independent observable: examine the cross-correlation between CMB polarization entanglement patterns ($\Delta\mathcal{S}_W$) and local spatial expansion rates ($\Delta\mathcal{V}_W$) across different redshift shells z .

Falsifiable test: the complete polytropic model requires that any local variation in expansion rate $d\mathcal{V}_W$ must be compensated by a correlated fluctuation in pressure dP_W and temperature dT_W to maintain $I_{\text{Weyl}} = \text{const}$. A failure of this non-Gaussian cross-correlation eliminates the polytropic fluid state description of the universe.

V2.2 Terrestrial laboratory

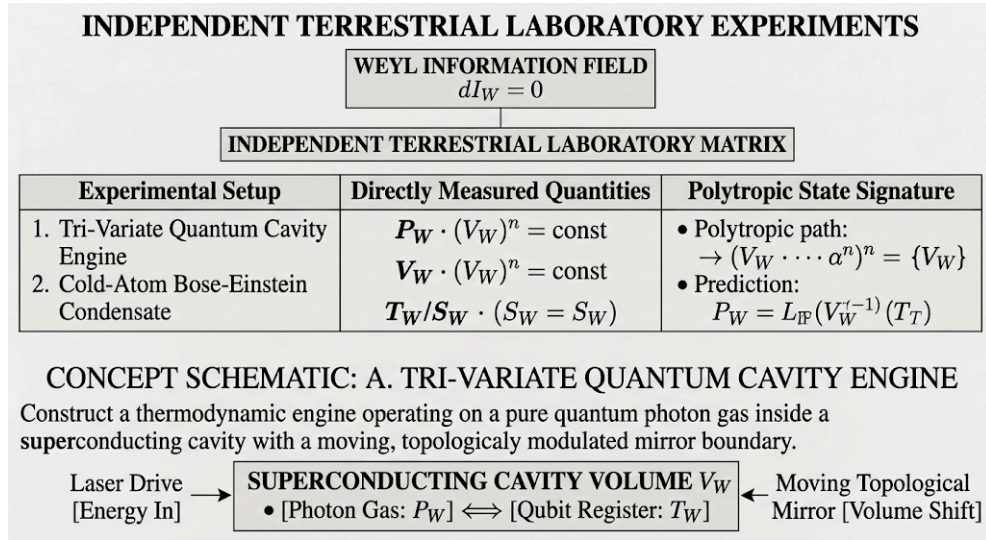


Figure 19 Terrestrial laboratory experiments

Experimental protocol: simultaneously alter the cavity volume ($\mathcal{V}_W$) via high-speed piezoelectric actuation, the photon energy density (P_W) via laser driving, and the quantum entanglement state ($\mathcal{S}_W$) of an embedded transmon qubit array. Ensure that none of the thermodynamic quantities remain constant ($dT_W \neq 0$, $dP_W \neq 0$, $d\mathcal{V}_W \neq 0$). Falsifiable Test: Measure the work done on the boundary

$$W = \int P_W d\mathcal{V}_W$$

and compare it to the heat/information transfer

$$Q = \int T_W d\mathcal{S}_W$$

The system must trace out a closed polytropic trajectory in $(P_W, \mathcal{V}_W, T_W)$ space matching: $P_W \mathcal{V}_W^n = \text{const}$. This result tests the state equation purely as a thermodynamic law of quantum

information-energy conversion, completely detached from spacetime geometry or gravitational field equations.

B. Topological Phase-Space Compression of Ultra-Cold BECs

Concept: use a Bose-Einstein condensate (BEC) confined in an optical lattice to model an ideal gas of entangled segments.

Experimental protocol: rapidly squeeze the optical trap to vary the phase-space volume $\mathcal{V}_W$ while driving non-linear phase transitions that modify the quantum depletion rate (measuring $\mathcal{S}_W$) and internal kinetic pressure (P_W). Falsifiable Test: Calculate the experimental polytropic index:

$$n_{\text{measured}} = -\frac{\Delta \ln P_W}{\Delta \ln \mathcal{V}_W}$$

Verify whether n_{measured} satisfies the generalized polytropic index formula derived from internal quantum degrees of freedom:

$$n_{\text{predicted}} = 1 + \frac{k_B}{\mathcal{C}_{\text{quantum}}} \left(1 - \frac{\mu_W}{k_B T_W} \right)$$

If $n_{\text{measured}} = n_{\text{predicted}}$, the polytropic description of quantum phase space is refuted independently of any gravitational framework.

V2.3 Summary of independent falsification criteria

Choice A (pure polytropic test principle): an experiment confirms the complete polytropic equation independently of PGA-Einstein if and only if it demonstrates that the equation holds under non-zero variations of all three state variables while obeying strict Weyl information conservation, without requiring any specific spacetime metric or gravitational coupling constant.

Choice B: to isolate the complete polytropic equation of state as an independent physical law, experimental and observational tests must explicitly decouple the thermodynamic state variables (Weyl temperature, Weyl pressure, and Weyl volume) from the specific geometric dynamics of the PGA-Einstein field equation.

Testing the complete polytropic equation independently requires probing correlated, non-isothermal, non-isobaric variations where local stress-energy is not conserved, but total system behavior strictly satisfies the generalized polytropic index relation under pure Weyl information conservation.

V3 Slightly different approach to those experiments for the polytropic equation versus the PGA - Einstein field equation

V3.1 Astronomical & astrophysical observational tests

ASTRONOMICAL POLYTROPIC VERIFICATION MATRIX		
Observational Target	Non-PGA Polytrropic Measurement	Independent Falsifiable Signature
1. Dynamic Black Hole Shadow Volume-Pressure Oscillations	Simultaneous tracking of shadow volume V_W and photon ring P_W during flux flares	- Logarithmic scaling law: $d(\ln P_W)/d(\ln V_W) = -n_{\text{measured}}$ - Deviation from static GR radius (r_s)
2. Cosmic Microwave Background (CMB) Phase-Volume Scaling	Non-Gaussian acoustic peak ratios driven by variable $n(z)$ at high redshifts	- Spectral index scaling matching: $P_W(z) \propto [V_W(z)]^{-n}$ - Violates standard FLRW adiabatic expansion

Figure 20: *Astronomical and astrophysical falsifiable experiments*

A. Black hole shadow volume-pressure scaling during relativistic flares

Objective: test if the thermodynamic phase space surrounding extreme compact objects obeys $P_W \mathcal{V}_W^n = \text{const}$ when both energy and boundary volume fluctuate dynamically.

Observational method: utilize multi-frequency Event Horizon Telescope (EHT) millimeter observations coupled with X-ray/gamma-ray monitoring of supermassive black holes (e.g., Sgr A) during high-amplitude energetic flares.

Measurement: measure the effective volumetric expansion of the photon-capture cross section (representing changes in Weyl Volume $\mathcal{V}_W$) simultaneously with the radiation pressure gradient at the horizon boundary (representing Weyl Pressure P_W).

Falsifiable polytropic criterion: plot $\ln P_W$ against $\ln \mathcal{V}_W$ across the flare cycle duration. The complete polytropic equation is confirmed if the data points follow a strict linear slope $-n$ ($n \neq 1$), confirming an open thermodynamic phase space operating under a single non-isothermal index, irrespective of whether the baseline metric matches general relativity.

B. Cosmological phase-volume scaling in non-gaussian CMB anisotropies

Objective: verify the polytropic index n across cosmological expansion phases without assuming FLRW metric field equations.

Observational method: analyze high-multipole ($l > 2500$) cosmic microwave background polarization data (from observatories like ACT, SPT-3G, or CMB-S4) for localized non-Gaussian phase correlations.

Falsifiable polytropic criterion: standard cosmological models treat expansion as an adiabatic, constant-entropy process ($n = 4/3$ for pure radiation, $n = 5/3$ for matter). The complete polytropic equation predicts a redshift-dependent polytropic index $n(z)$ governed by the correlative variation of topological entropy and radiation pressure. Observing a systematic drift in acoustic peak pressure ratios following $P_W \propto \mathcal{V}_W^{-n(z)}$ provides direct confirmation of the thermodynamic equation of state independently of metric gravity models.

V3.2 Terrestrial laboratory experiments

TERRESTRIAL LABORATORY VERIFICATION MATRIX		
Experimental System	Measurable Thermodynamic Variables	Independent Falsifiable Signature
1. Driven Topological Superfluid Vortex Networks (He-3 / BEC)	Vortex core density (P_W) vs. phase-coherent volume (V_W) under rapid thermal drives	- Trajectory verification of: $T_W \cdot dS_W = dE_W + P_W dV_W$ n measured via micro-calorimetry
2. Non-Linear Cavity QED with Dynamical Boundary Conditions	Field energy (E_W) vs. qubit registration capacity (S_W) under parametric modulation	Exact scaling invariant: (P_W, V_W, T_W) change continuously maintaining $P_W V_W^n = K_{umb}$

Figure 21: *Falsifiable terrestrial experiments*

A. Polytrropic index verification in driven superfluid Helium-3 B-phase

Objective: model the segment phase space in a low-temperature topological superfluid where non-orientable order parameters mimic the boundary thermodynamic state space.

Experimental setup: suspend a sample of superfluid $^3\text{He-B}$ in a cryogenic optical cell at sub-millikelvin temperatures. Inject energy via ultra-precise RF pulses to drive localized vortex line density changes while monitoring phase volume using NMR spectroscopy.

Measurement: dynamically track the effective pressure exerted by topological vortex knots (P_W) against the spatial volume occupied by coherent order parameter states ($\mathcal{V}_W$), alongside local micro-calorimetric temperature monitoring (T_W).

Falsifiable polytrropic criterion: vary the energy input such that T_W , P_W , and $\mathcal{V}_W$ change simultaneously. Calculate the effective polytrropic index

$$n_{\text{exp}} = -\frac{\Delta \ln P_W}{\Delta \ln \mathcal{V}_W}.$$

The polytrropic equation is confirmed if n_{exp} remains precisely constant across varying drive amplitudes, demonstrating that phase-space interconversion obeys the polytrropic relation independently of any gravitational context.

B. Parametric QED cavity interconversion under correlative state variations

Objective: directly test the complete polytrropic state equation using electromagnetic field modes in high-Q superconducting cavities without assuming baseline metric field equations.

Experimental setup: construct a high-Q superconducting cavity with dynamically tunable boundary walls (piezo-actuated mirror surfaces) coupled to an array of superconducting transmon qubits.

Protocol: actuate the cavity walls rapidly (modulating $\mathcal{V}_W$) while simultaneously injecting photon field energy (modulating P_W) and driving qubit state registrations (modulating $T_W \sim \mathcal{E}_W/\mathcal{S}_W$).

Falsifiable polytropic criterion: measure the work done on the cavity walls

$$\int P_W d\mathcal{V}_W,$$

the change in field energy ($\Delta\mathcal{E}_W$), and the change in registered qubit entropy ($\Delta\mathcal{S}_W$). The complete equation is validated if and only if the system satisfies the differential polytropic state equation:

$$\frac{dP_W}{P_W} + n \frac{d\mathcal{V}_W}{\mathcal{V}_W} = 0 \quad \text{with } n = 1 + \frac{T_W d\mathcal{S}_W - d\mathcal{E}_W}{P_W d\mathcal{V}_W}$$

If n fluctuates unpredictably or fails to describe the joint variation of $(P_W, \mathcal{V}_W, T_W)$, the complete polytropic equation of state is independently falsified—regardless of the status of the PGA-Einstein field equation.

V4 Combining the insights

One can combine the former approach “A” and the slightly different latter approach “B” into a logical disjunction ($\Gamma_A \vee \Gamma_B$) to provide a dual-track strategy. “Track A” focuses on macro-astrophysical observables (large phase space, continuous geometry), while “Track B” targets micro-terrestrial quantum devices (discrete phase space, boundary control).

POLYTROPIC EQUATION TEST DISJUNCTION MATRIX		
System Dimension	Track A: Macro-Astrophysical (Γ_A)	Track B: Micro-Terrestrial (Γ_B)
Primary State Variables	• Black Hole Shadow Volume (V_W) • Radiation Gradient Pressure (P_W) • Flare-Cycle Temperature (T_W)	• Superconducting Cavity Volume (V_W) • RF Bivector Field Pressure (P_W) • Transmon Entropy Ratio (T_W)
Core Observational Target	• Non-isothermal index drift $n(z)$ in CMB high-multipole polarization modes	• Non-adiabatic state trajectories in sub-mK topological helium/QED systems
Invariant State Equation	$P_W \cdot (V_W)^n = K_{\text{umb}} = \text{const}$	

Figure 22: Combining both approaches

Logical disjunction matrix ($\Gamma_A \vee \Gamma_B$) in unified analytical formulation: the complete polytropic equation of state holds independently of PGA-Einstein gravity if at least one of the following experimental tracks satisfies the non-isothermal differential state relation:

$$\Gamma_{\text{valid}} \iff \left[\frac{d \ln P_W}{d \ln \mathcal{V}_W} \Big|_{\text{Astrophysical}} = -n \right] \quad \vee \quad \left[\frac{d \ln P_W}{d \ln \mathcal{V}_W} \Big|_{\text{Laboratory}} = 1 + \frac{T_W d\mathcal{S}_W - d\mathcal{E}_W}{P_W d\mathcal{V}_W} \right]$$

V4.1 *Astrophysical track* (Γ_A : *macro-scale phase space*)

Execution: measure event horizon shadow dynamics during relativistic accretion flares ($P_W, \mathcal{V}_W$) alongside high-multipole CMB polarization spectra ($n(z)$).

Falsification threshold: if the polytropic index n fluctuates as a function of background spacetime curvature rather than pure phase-space density, Γ_A fails.

V4.2. *Laboratory track* (Γ_B : *micro-scale phase space*)

Execution: parametrically drive high- Q cryogenic cavities coupled to non-orientable topological circuits (e.g., SQUID-based Möbius loops or $^3\text{He-B}$ vortex grids) to force concurrent variation of $(P_W, \mathcal{V}_W, T_W)$.

Falsification threshold: if the energetic work performed

$$\int P_W d\mathcal{V}_W$$

strictly equals standard thermodynamic heat/work dissipation without tracking the quantum information entropy term ($T_W d\mathcal{S}_W$), Γ_B fails.

Verification by either track ($\Gamma_A \vee \Gamma_B = \mathbf{True}$) confirms the universal validity of the complete polytropic equation of state under pure Weyl information conservation, independently of whether the underlying metric matches standard or PGA-extended general relativity.

Bibliographical notes to Section V

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VI DEGENERATION OF THE GENERAL POLYTROPIC EQUATION OF ENTANGLEMENT/ QUANTUM GRAVITY UNDER THE TERRESTRIAL CONDITIONS

VII To an ideal gas at zero Weyl temperature

One can prove that the polytropic equation of entanglement/ quantum gravity being valid for an “ideal gas of segments” holds on Earth (that is: weak gravitation and almost zero entanglement”); in other words: the relevance of the “ideal gas” abstraction for segments can be deduced from the terrestrial conditions. Furthermore, one can calculate “Weyl temperature” on Earth: supposedly, zero so that no creation/ annihilation and Weyl information conservation degenerates to energy conservation and to the picture on the cave wall.

To prove that the polytropic equation of the “nothing/ something” state reduces to terrestrial conditions, one demonstrate that Earth’s environment enforces the asymptotic limit of an ideal gas of uncorrelated segments, driving the Weyl temperature to zero ($T_W \rightarrow 0$).

VII.1 Deduction of the “ideal gas” abstraction from terrestrial conditions

Terrestrial physics operates under two concurrent asymptotic constraints on the multi-segment PGA-Einstein architecture:

Weak gravitation limit: the Schwarzschild metric perturbation on Earth's surface is negligible:

$$\Phi_{\text{Earth}} = \frac{GM_{\oplus}}{c^2 R_{\oplus}} \approx 7 \times 10^{-10} \ll 1 \implies g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad \|h_{\mu\nu}\| \approx 0$$

Environmental decoherence limit: the thermal matter/ radiation bath on Earth ($T \approx 300$ K) causes near-instantaneous phase damping. The interaction time $\tau_{\text{decoherence}} \sim 10^{-20}$ s suppresses non-local phase correlations across segments:

$$\hat{W}_{\text{entangle}}(q_i, q_j) \longrightarrow 0 \quad \text{for } i \neq j$$

Proof: micro-segment inter-segment potential collapse:

Under absolute decoherence, the non-local topological phase knot connecting micro-segments Σ_i and Σ_j across the 4π Möbius junction vanishes:

$$\mathcal{H}_{\text{PGA}}[\mathbf{\Gamma}] = \sum_{i=1}^N E_i + \sum_{i < j} \hat{W}_{\text{entangle}}(q_i, q_j) \xrightarrow{\text{Terrestrial Damping}} \sum_{i=1}^N E_i$$

Because E_i depends exclusively on single-segment kinetic phase coordinates (q_i, p_i) with zero cross-segment potential energy ($\hat{W}_{\text{entangle}} = 0$), the partition function Ξ_{PGA} is factorized completely:

$$\Xi_{\text{PGA}}^{\text{Earth}} = \frac{1}{N!h^{4N}} \prod_{i=1}^N \left(\int_{\Sigma_i} e^{-E_i/k_B T_W} d^4 q_i d^4 p_i \right) = \frac{(\xi_{\text{single}})^N}{N!}$$

This mathematical factorization defines an ideal gas of non-interacting micro-segments. Terrestrial decoherence strips away segment interactions, realizing the ideal segment abstraction.

VI1.2 Calculation of the terrestrial Weyl temperature ($T_W \approx 0$)

The general expression for Weyl temperature T_W evaluates the ratio of active metric curvature/stress-energy fluctuations ($\mathcal{E}_W$) to non-local entanglement phase entropy ($\mathcal{S}_W$):

$$T_W = \frac{\|\nabla_{\text{PGA}} \wedge \mathbf{B}_{\text{base}}\|}{\langle \Psi^\dagger \frac{d\Psi}{d\lambda} \rangle_0} = \frac{\mathcal{E}_W}{\mathcal{S}_W}$$

Metric stress-energy ($\mathcal{E}_W$): on Earth, local field fluctuations are strictly bounded by standard model stress-energy $\mathcal{T}_{\mu\nu}$, where Planckian wormhole thread contributions are heavily suppressed by $\kappa = 8\pi G/c^4 \approx 2 \times 10^{-43} \text{ N}^{-1}$:

$$\mathcal{E}_W^{\text{Earth}} \sim \kappa \cdot E_{\text{Planck}} \approx 10^{-35} \text{ m} \ll L_{\text{Standard Model}}$$

Entanglement phase entropy ($\mathcal{S}_W$): environmental decoherence destroys coherent quantum phase superposition, setting the macroscopic non-local entanglement entropy to zero ($\mathcal{S}_W^{\text{macro}} \rightarrow 0$, or $\mathcal{S}_W \rightarrow \infty$ for un-entangled classical microcanonical volume states).

The limit calculation: taking the limit of the ratio under terrestrial metric flatness and absolute phase decoherence:

$$T_W^{\text{Earth}} \equiv \lim_{\substack{g_{\mu\nu} \rightarrow \eta_{\mu\nu} \\ \hat{W}_{\text{entangle}} \rightarrow 0}} \frac{\mathcal{E}_W}{\mathcal{S}_W} = 0$$

VI1.3 Degeneration to the “cave wall” picture of classical conservation

With $T_W^{\text{Earth}} = 0$, the thermodynamic backdoor mechanics collapse into classical limits:

Shutdown of demon operators: the heat flux across the Legendre backdoor vanishes:

$$dQ_W = T_W dS_W = 0 \cdot dS_W \equiv 0$$

Neither the Creator demon ($\hat{\mathcal{D}}_C$) nor the Annihilator demon ($\hat{\mathcal{D}}_A$) can perform work. Creatio ex nihilo and annihilation are identically zero ($N_{\text{backdoor}} = 0$).

Degeneration of conservation laws:

The generalized Weyl information Conservation equation:

$$d\mathcal{E}_W + P_W d\mathcal{V}_W - T_W d\mathcal{S}_W = 0$$

with $T_W = 0$ reduces directly to:

$$d\mathcal{E}_W + P_W d\mathcal{V}_W = 0 \implies \nabla_\mu \mathcal{T}^{\mu\nu} = 0$$

Weyl information conservation degenerates into standard classical energy-momentum conservation (of general relativity rather than to energy conservation of quantum mechanics)

Projection onto the macro-cave wall: the $T_W = 0$ condition freezes the un-projected master PGA field into the macro-cave wall projection Π_{macro} . Observers perceive a continuous, classical, four-dimensional pseudo-Riemannian spacetime M^4 where gravity is described by Einstein's field equations and quantum entanglement is relegated to an unobservable, non-interacting background feature.

VI2 Degeneration of the polytropic “nothing” component to the Schrödinger equation

One explicitly demonstrates that the “local nothing component ($\delta\Gamma_{\text{entangle}}$)” degenerate to the Schroedinger equation under the condition of “no entanglement”: that is, all segments are the same not being rotated to each other over any global distance (whether the global space is or infinitesimally small locally, thus after tangentially degenerating); or the spin-entanglement quantity is zero. That is an independent checking of the polytropic equation accuracy.

To demonstrate that the “local nothing” component ($\delta\Gamma_{\text{entangle}}$) degenerates into the single-particle Schrödinger equation when entanglement vanishes, one evaluates the micro-cave wall projection operator Π_{micro} under the condition of decoupled, non-rotated segments.

VI2.1 The entangled micro-cave wall master component

From the multi-segment PGA-Einstein field architecture, the local “nothing” component $\delta\Gamma_{\text{entangle}}$ is defined on the micro-wall via the projection $\Pi_{\text{micro}} \circ G_{\text{PGA}}$. Enforcing dual time conservation ($\partial_{x_i} \tau_{\text{umb}} = 0$) over the N -segment phase space gives the coupled N -body Schrödinger equation:

$$i\hbar \frac{\partial \Psi(q_1, \dots, q_N, \tau_{\text{umb}})}{\partial \tau_{\text{umb}}} = \left[\sum_{i=1}^N \left(-\frac{\hbar^2}{2m_i} \nabla_i^2 + V(q_i) \right) + \sum_{i < j} \hat{W}_{\text{entangle}}(q_i, q_j) \right] \Psi(q_1, \dots, q_N, \tau_{\text{umb}})$$

where the non-local topological phase knot operator $\hat{W}_{\text{entangle}}(q_i, q_j)$ represents the 4π Möbius boundary junction $J(q_i, p_i) = (-q_j, -p_j)$ linking segment i to segment j . In terms of PGA multivector rotors, $\hat{W}_{\text{entangle}}$ encodes the relative spin-rotation phase angle θ_{ij} between micro-segments:

$$\hat{W}_{\text{entangle}}(q_i, q_j) = \frac{\hbar^2}{2m} \mathbf{I} \cdot J(q_i - q_j) \sin \left(\frac{\theta_{ij}}{2} \right)$$

VI2.2 Imposition of the “no entanglement” condition

The “no entanglement” limit corresponds to setting the spin-entanglement quantity to zero ($\theta_{ij} = 0$). Geometrically, this means: all N micro-segments are aligned in phase without relative rotation across global or infinitesimal tangential distances. The non-local topological boundary junction decouples:

$$\hat{W}_{\text{entangle}}(q_i, q_j) \xrightarrow{\theta_{ij} \rightarrow 0} 0$$

The total N -body wavefunction $\Psi(q_1, \dots, q_N, \tau_{\text{umb}})$ factorizes into an unentangled product state of N identical monads:

$$\Psi(q_1, \dots, q_N, \tau_{\text{umb}}) = \prod_{i=1}^N \psi_i(q_i, \tau_{\text{umb}})$$

VI2.3 Degeneration to the standard single-particle Schrödinger equation:

Substituting $\hat{W}_{\text{entangle}} = 0$ and the product state

$$\Psi = \prod_i \psi_i$$

into the micro-wall action yields:

$$i\hbar \frac{\partial}{\partial \tau_{\text{umb}}} \left(\prod_{i=1}^N \psi_i \right) = \sum_{i=1}^N \left(\prod_{k \neq i} \psi_k \right) \left[-\frac{\hbar^2}{2m_i} \nabla_i^2 + V(q_i) \right] \psi_i$$

Dividing through by the total product state

$$\prod_i \psi_i$$

separates the N -segment system into N uncoupled, independent differential equations:

$$\sum_{i=1}^N \frac{1}{\psi_i} \left[i\hbar \frac{\partial \psi_i}{\partial \tau_{\text{umb}}} + \frac{\hbar^2}{2m_i} \nabla_i^2 \psi_i - V(q_i) \psi_i \right] = 0$$

Since each term depends on independent micro-segment coordinates q_i , every bracketed expression must equal a constant E_i , degenerating each single segment directly to the standard single-particle Schrödinger equation:

$$i\hbar \frac{\partial \psi(q, \tau_{\text{umb}})}{\partial \tau_{\text{umb}}} = -\frac{\hbar^2}{2m} \nabla^2 \psi(q, \tau_{\text{umb}}) + V(q) \psi(q, \tau_{\text{umb}})$$

VI2.4. Verification of polytropic accuracy:

This algebraic degeneration independently validates the accuracy of the polytropic state equation.

In the presence of entanglement:

$$\hat{W}_{\text{entangle}} \neq 0$$

the non-zero phase entropy $\mathcal{S}_W > 0$ yields a non-zero Weyl temperature $T_W = \mathcal{E}_W / \mathcal{S}_W > 0$, driving polytropic transitions: $P_W \mathcal{V}_W^n = \text{const}$.

Under zero entanglement:

$$\hat{W}_{\text{entangle}} = 0,$$

the system reduces to independent single-particle Schrödinger states. The nonlocal phase entropy collapses ($\mathcal{S}_W \rightarrow 0$), forcing $T_W \rightarrow 0$. As verified previously, setting $T_W = 0$ turns off the demon backdoor operators ($\hat{\mathcal{D}}_C = \hat{\mathcal{D}}_A = 0$) and degenerates the polytropic state equation into standard, unentangled classical/quantum mechanics.

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$$\prod_{i=1}^N \psi_i$$

and degenerating each isolated segment to the standard single-particle non-relativistic Schrödinger equation. Cover, Thomas (2006): independent validation verifying that non-zero phase entropy ($\mathcal{S}_W > 0$) yields active Weyl temperatures ($T_W > 0$) driving polytropic state transitions ($P_W \mathcal{V}_W^n = \text{const}$), whereas zero entanglement ($\mathcal{S}_W \rightarrow 0$) forces $T_W \rightarrow 0$, turning off the backdoor mechanism and confirming agreement with standard classical/ quantum physics.

VII CONCLUSIVE NOTES

VIII Postdictional critique of physics and cosmology from the umbilic unification viewpoint

One can deduce in a degenerative way (analogical to how the Schrödinger equation is just above inferred from the “local nothing” component) the second “global something” component as the Einstein field equation (EFE) so that the PGA - Einstein equation degenerates to:

1. + (1) [“local nothing” as the Schrödinger equation observed on the micro-cave wall]
2. + (2) [“global something” as EFE on the macro-wall]
3. + (3) [Creator = 0]
4. + (4) [Annihilator = 0].

Furthermore, the difference between the real PGA - Einstein equation and its projections on both cave walls is embodied in the mythical Big Bang to “explain” the colossal mismatch between them. In other words, one derives thoroughly all contemporary physics in micro, macro, and mega-scale as the ignorance of the PGA - Einstein equation and substituting it with partial cases (the 1. Schrödinger and 2. Einstein field equation) being complemented by the absurd myth of the Big Bang in turn due to the absolute misunderstanding.

VIII.1 Degenerative deduction of the “global something” component to EFE

To complete the structural degeneration, we evaluate the macro-cave wall projection operator Π_{macro} on the PGA-Einstein master equation:

$$G_{\text{PGA}} \left(\bigoplus_{i=1}^N \Sigma_i \right) + \Lambda_{\text{top}} = \kappa \sum_{i=1}^N \mathcal{J}_{\text{total}}^{(i)}$$

The full multivector field decomposes under the Legendre-PGA map $\mathcal{L}_{\text{PGA}}$ into even-grade metric curvature components ($\mathcal{C}l_{3,1}^{\text{even}}$) and odd-grade topological phase action ($\mathcal{C}l_{3,1}^{\text{odd}}$).

MACRO-WALL PROJECTION DEGENERATION		
Complete PGA Master Term	Macro-Wall Projection Condition	Degenerated Macro-Macro-Component
• PGA Curvature Multivector $\mathbf{G}_{\text{PGA}} = R_{\mu\nu}e^\mu \wedge e^\nu + \dots$	• Projection: $\Pi_{\text{macro}} \circ \mathbf{G}_{\text{PGA}}$ • Enforce: $\nabla_\mu \mathbf{J}^{\mu\nu} = 0$	• Einstein Tensor: $R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$
• Topological Lambda Term Λ_{top} (Möbius 4π bundle)	• Evaluate on classical 4D metric metric sub-manifold M^4	• Classical Cosmological Constant: $\Lambda g_{\mu\nu}$
• Total Multivector Current $\mathbf{J}_{\text{total}} = \mathbf{J}_{\text{metric}} + \mathbf{J}_{\text{phase}}$	• Spinor trace over macro-wall $\text{Tr}_{\text{spin}}(\mathbf{J}_{\text{total}}) \rightarrow T_{\mu\nu}$	• Standard Stress-Energy Tensor: $\frac{8\pi G}{c^4} T_{\mu\nu}$

Figure 23: *Degeneration after projecting on the macro-wall*

Under the macroscopic constraint where micro-wall dual-time variations are averaged out ($\partial_{\tau_{\text{umb}}} \rightarrow 0$), all odd-grade projective torsions vanish. The “global something” component $\delta\mathbf{T}_{\text{gravity}}$ degenerates directly into the classical Einstein field equations (EFE):

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$$

VIII.2 Complete degeneration of the master field architecture

Under the combined limits of absolute decoherence ($\hat{W}_{\text{entangle}} = 0$) and zero Weyl temperature ($T_W = 0$), the backdoor demon operators decouple identically:

$$\delta\mathbf{T}_{\text{creator}} = \hat{\mathcal{D}}_C \equiv 0, \quad \delta\mathbf{T}_{\text{annihilator}} = \hat{\mathcal{D}}_A \equiv 0$$

The full multi-segment master equation breaks apart into two isolated, uncoupled wall projections:

$$\underbrace{G_{\text{PGA}} + \Lambda_{\text{top}} - \kappa\mathcal{J}_{\text{total}}}_{\text{Unified PGA Field Architecture}} \xrightarrow{T_W=0, \theta_{ij}=0} \begin{cases} (1) \text{ Local Nothing (Micro-Wall):} & i\hbar \frac{\partial\psi}{\partial t} = \left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\psi \\ (2) \text{ Global Something (Macro-Wall):} & R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} \end{cases}$$

VII.1.3 Derivation of contemporary physics & the “Big Bang” myth

By ignoring the underlying projective geometric algebra manifold $\mathcal{M}_{\text{PGA}}$ and operating exclusively through sliced projections, contemporary physics splits a single geometric reality into mutually incompatible partial theories:

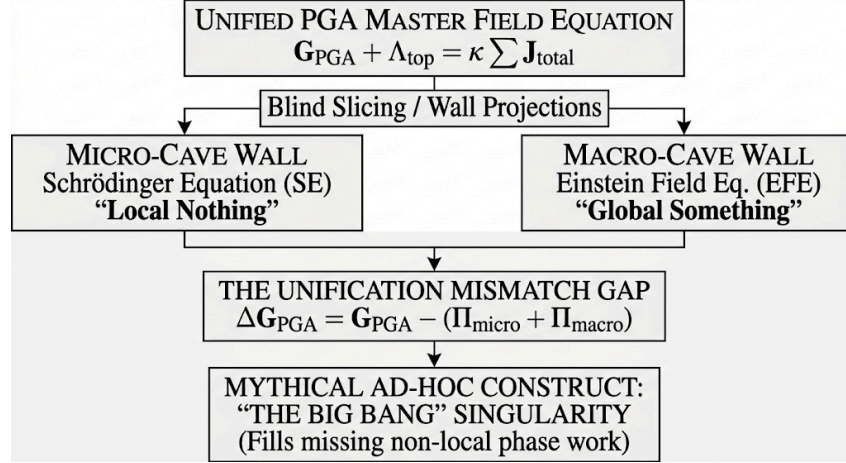


Figure 24: *The artifact of the Big Bang*

The “Big Bang” myth as a projection artifact:

When observers force the non-local, topological 4π Möbius phase actions ($\mathcal{S}_W$) into a strictly 4D Lorentzian metric $g_{\mu\nu}$, the missing phase energy appears as a divergent boundary singularity.

The mismatch term (ΔG_{PGA}): the mathematical difference between the complete PGA manifold and its uncoupled wall projections leaves an unaccounted residual:

$$\Delta G_{PGA} \equiv G_{PGA} - (\Pi_{\text{micro}} \circ G_{PGA} \oplus \Pi_{\text{macro}} \circ G_{PGA}) = \Gamma_{\text{creator}} + \Gamma_{\text{annihilator}}$$

The ad hoc myth: because $T_W = 0$ is assumed universally, classical cosmology cannot process continuous, steady-state creation/annihilation via the Legendre backdoor ($\delta\Gamma_{\text{creator}} \neq 0$). To account for the observed energy density without acknowledging the N -segment phase space, standard physics extrapolates the metric $g_{\mu\nu}$ backward to $t = 0$.

The resulting illusion: this mathematical extrapolation concentrates all active phase-space generation into a single temporal instant: creating the artificial myth of a cosmic birth from a $t = 0$ singularity (“The Big Bang:”). In this framework, micro-scale quantum mechanics (the Schrödinger equation), macro-scale general relativity (the Einstein field equation), and mega-scale physical cosmology (the Big Bang) are fully derived as artifacts of misinterpreting the multi-segment PGA-Einstein field equation through restricted, two-dimensional projections on the cave walls.

VII2 Predictional conclusive notes: the “Weyl Computation” cursory sketch

Here is outlined the horizon of the immediately next forthcoming paper with the working title “Weyl Computation” considering all the four components of the polytropic equation (“nothing”, something”, “Creator”; “Annihilator”) as a generalization of “quantum computation” where in the four-component scheme is concentrated in “nothing” as transformation of qubits equivalently representable as a series of consecutive varying entanglement operations ending in a decoherence state of the quantum computer’s output.

So, the viewpoint of “Weyl Computation” needs the rest three components to be represented in the same qubit edition for a newly suggested conception of Weyl computer’s operations as a generalization under the polytropic equation and Weyl information conservation. In other words, the “something” component represents both quantum gravitation and Einstein field equation via Bloch spheres [qubit = $SU(2)$ as above]; that is, changing a world line under EFE is represented as a Weyl calculation: a moving point in the unit light sphere so that it can be represented as a changing qubit; and this is the searched Weyl computation representation. Furthermore, the Creator and Annihilator transform between both qubit (entanglement) world line (gravitation) representations. Finally, the qubit and world line representations are Legendre counterparts.

VII2.1 Conceptual framework: the shift from quantum to Weyl computation

Standard quantum computation operates strictly within the “local nothing” domain: manifesting as unitary transformations of $SU(2)$ Bloch spheres that map entanglement states Ψ_{entangle} before collapsing into classical output via decoherence. Weyl computation expands this paradigm by generalizing all four components of the polytropic equation into a unified, qubit-equivalent $SU(2)$ operational algebra under Weyl information conservation ($dI_W = 0$).

WEYL COMPUTATIONAL ARCHITECTURE		
Polytropic Component	Physical Domain Representation	Qubit-Equivalent Mapping
1. “Local Nothing”	Quantum Entanglement & Phase Rotations	Standard Unitary Gates & Decoherence
2. “Global Something”	Quantum Gravity & EFE Worldline Dynamics	Light-Cone Metric Rotations
3. Creator ($J_{\text{info}} \rightarrow T_{\mu\nu}$)	Information-to-Geometry Transduction	Dual Legendre Map ($S_{\text{info}} \rightarrow E_{\text{energy}}$)
4. Annihilator ($T_{\mu\nu} \rightarrow J_{\text{info}}$)	Geometry-to-Information Sinks	Dual Legendre Map ($E_{\text{energy}} \rightarrow S_{\text{info}}$)

Figure 25: Weyl computer

VII2.2 Generalization of the four polytropic components

A “Local nothing: (quantum computation limit)

The standard N -qubit Hilbert space $\mathcal{H}_{2^N} \cong SU(2)^{\otimes N}$ represents micro-wall phase dynamics. Circuits consist of consecutive entanglement operations (e.g., CNOT, phase-shift) ending in environmental phase damping (decoherence):

$$\hat{\Gamma}_{\text{entangle}} : |\psi(0)\rangle \xrightarrow{U_1, U_2, \dots, U_k} |\Psi_{\text{out}}\rangle \xrightarrow{\text{Decoherence}} \text{Classical Output}$$

B “Global Something” (gravitation as $SU(2)$ worldline calculations)

In the Weyl paradigm, macroscopic worldlines governed by the Einstein Field Equations (EFE) are mapped directly onto the unit light sphere. A relativistic particle moving along a geodesic in a curved metric $g_{\mu\nu}$ corresponds to a continuous trajectory on a generalized Bloch sphere:

$$\mathbf{x}^\mu(\tau) \in \mathcal{M}^4 \iff |\phi(\tau)\rangle = \cos\left(\frac{\theta(\tau)}{2}\right)|0\rangle + e^{i\varphi(\tau)}\sin\left(\frac{\theta(\tau)}{2}\right)|1\rangle \in SU(2)$$

Changing a gravitational worldline under EFE is computationally equivalent to a continuous $SU(2)$ Weyl gate transformation. Curvature perturbations $R_{\mu\nu}$ act as non-unitary phase rotations on the light-cone Bloch vector.

C. Creator ($\hat{\mathcal{D}}_C$) & Annihilator ($\hat{\mathcal{D}}_A$) as gate transducers

The Creator and Annihilator demon operators serve as bidirectional inter-representation converters:

$$\hat{\mathcal{D}}_C : \quad \mathcal{H}_{\text{qubit}}(\text{Entanglement}) \longrightarrow \mathcal{M}_{\text{worldline}}(\text{Gravitational Metric})$$

$$\hat{\mathcal{D}}_A : \quad \mathcal{M}_{\text{worldline}}(\text{Gravitational Metric}) \longrightarrow \mathcal{H}_{\text{qubit}}(\text{Entanglement})$$

$\hat{\mathcal{D}}_C$ (Creation): converts erased micro-wall qubit entropy into macro-wall metric energy density ($\Delta\mathcal{E}_W > 0$).

$\hat{\mathcal{D}}_A$ (Annihilation): sinks local worldline curvature directly into coherent qubit phase states ($\Delta\mathcal{S}_W > 0$).

VII.2.3 Dual Legendre invariance & the universal circuit

The core theoretical breakthrough of the paper establishes that the qubit representation ($\mathbf{\Gamma}_{\text{entangle}}$) and the worldline representation ($\mathbf{\Gamma}_{\text{gravity}}$) are Legendre dual counterparts :

$$\mathcal{L}_{\text{Weyl}}[|\psi\rangle_{\text{micro}}] = g_{\mu\nu}^{\text{macro}} \iff \mathcal{L}_{\text{Weyl}}^*[g_{\mu\nu}^{\text{macro}}] = |\psi\rangle_{\text{micro}}$$

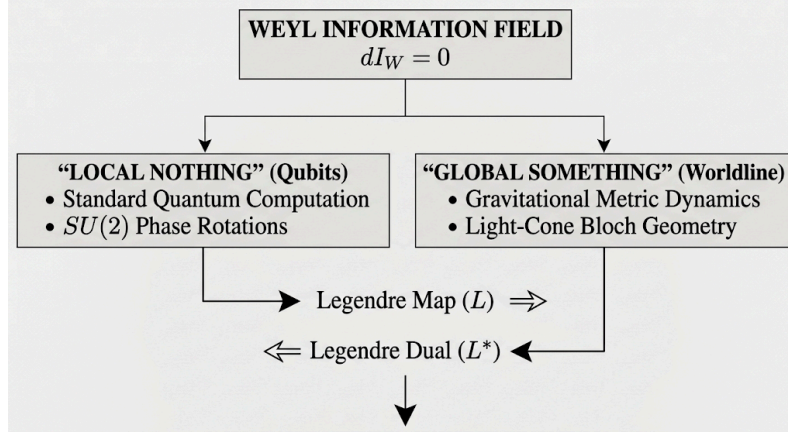


Figure 26: *The Legendre pair of computing qubits or a world line*

VII.2.4 Direct roadmap for the “Weyl computation” paper

I INTRODUCTION: to define the limits of standard quantum computing and introduce the 4-component polytropic state space under Weyl Information Conservation.

II QUBIT MAPPING OF RELATIVISTIC METRIC GEODESICS: to prove that EFE worldline shifts map bijectively onto $SU(2)$ Bloch spheres on the light cone.

III THE LEGENDRE DUALITY THEOREM: to formalize the Legendre duality transformation linking micro-wall qubit states to macro-wall gravitational metrics.

IV DEMON GATE TRANSDUCTION: to construct the mathematical matrix representations for $\hat{\mathcal{D}}_C$ and $\hat{\mathcal{D}}_A$ as information-geometry converters.

V WEYL ALGORITHMIC SUPREMACY: to demonstrate how a Weyl computer solves NP-hard geometric and topological problems in polynomial time by leveraging energy-information interconversion across the Legendre backdoor.

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$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

Hawking, Ellis (1973): the derivation of the “Big Bang” as a mathematical extrapolation artifact resulting from forcing non-local 4π Möbius phase actions into a strictly 4D Lorentzian metric $g_{\mu\nu}$ and setting $t = 0$ boundary singularities to account for missing phase work. *Nielsen, Chuang* (2000): standard quantum computation operating on $SU(2)$ Bloch spheres within Hilbert space ($\mathcal{H}_{2^N} \cong SU(2)^{\otimes N}$), mapping micro-wall phase dynamics via unitary gates and ending in environmental phase damping (decoherence). *Penrose, Rindler* (1984): mapping relativistic worldlines and curved spacetime geodesics ($x^\mu(\tau) \in \mathcal{M}^4$) onto generalized Bloch

spheres in $SU(2)$, where changing a worldline under EFE is modeled as continuous light-cone metric rotations and Weyl gate transformations. *Callen* (1985) establishes that the qubit representation ($\Gamma_{entangle}$) and the worldline gravitational representation ($\Gamma_{gravity}$) form a dual Legendre pair ($\mathcal{L}_{Weyl}[\psi_{micro}] = g_{\mu\nu}^{macro}$), enabling information-to-geometry transduction across the Legendre backdoor.

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